Peering into the twilight - a systematic review on visual-based investigation of mesophotic ecosystems - TEMPERATE

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DATA PREPARATION

Load libraries

```
library(here)

## here() starts at C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analys
is/Projects/Mesophotic_systematic_lit_rev_analysis

library(dplyr)

##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(readxl)   # to import excel data sets
library(magrittr)
library(stringr) #many data wrangling functions
library(readxl)
```

Load data

```
temperate <- read_excel(here("Extracted_data_analysis","Input_data","extracte
d_raw.xlsx"))
```

Subset for studies encompassing temperate latitudinal region

```
temperate <- subset(temperate, str_detect(temperate$latitudinalRegion, "tempe
rate"))
```

Cleaning

Clean year columns

```
#keep the year only in the year column (in some rows exact time is specified)
temperate$publicationYear <-  stringr::str_remove(temperate$publicationYear,
                                pattern = "\\-.*")
```

MISSING / ND / NOTAPP VALUES

Create a summary table

```
library(purrr)
```

```
##
## Attaching package: 'purrr'
```

```
## The following object is masked from 'package:magrittr':
##
##     set_names
```

```
library(tibble)

# Get all column names
cols <- names(temperate[,-c(1:13)])

# Create a summary list
summary_list <- map(cols, function(col) {
```

```
  data <- temperate[,-c(1:13)][[col]]

  tibble(
    Column     = col,
    unique     = paste(unique(data), collapse = ", "),
    nd         = sum(data == "nd", na.rm = TRUE),
    notapp     = sum(data == "notapp", na.rm = TRUE),
    assigned   = sum(!is.na(data) & data != "nd" & data != "notapp"),
    `NA`         = sum(is.na(data)),
    total      = sum(data == "nd", na.rm = TRUE) +
                 sum(data == "notapp", na.rm = TRUE) +
                 sum(!is.na(data) & data != "nd" & data != "notapp")
  )
})

# Combine into one dataframe
summary_df <- bind_rows(summary_list)

summary_df

## # A tibble: 72 × 7
##    Column                   unique                    nd notapp assigned  `NA`
 total
##    <chr>                    <chr>                  <int>  <int>    <int> <int>
 <int>
##  1 meso_direct_indirect     indirect, direct, m…       2      0      540     0
   542
##  2 meso_samplingPlatform    rov+trawl, acou+dre…       2      0      540     0
   542
##  3 taxonomic_publication    NA, taxa++, taxa           0      0       16   526
    16
##  4 meso_substrate           artificial+hard+sof…      12      0      497    33
   509
##  5 meso_habitat             cwc+mud+sandy+wreck…      10      0      494    38
   504
##  6 meso_protection          nd, yes, mixed, no       291      0      251     0
   542
##  7 taxonomicTargetLevel     group, comm+group, …       0     16      526     0
   542
##  8 habit                    sessile, mixed, not…       0     15      527     0
   542
##  9 environmentalPosition    endo+epi, epi, dem+…       0      7      535     0
   542
## 10 detected_meso_seagrass NA, x, notapp              0     10       26   506
    36
## # ℹ 62 more rows

rm(summary_list, cols)
```

GEOGRAPHICAL DISTRIBUTION

```
location <- temperate[,c (
  "recordTitle","DOI","publicationYear","prov_code","ecoreg_code")]
```

Load Marine Ecoregion of the World data

```
meow <- read.csv(file=here("Extracted_data_analysis","Input_data","MEOW.csv"))
```

Provinces

```
prov <- location

# extract all unique province levels
prov_lvls <- unique(unlist(str_split(location$prov_code, "_")))

# create columns with exact matching
prov[, prov_lvls] <- t(sapply(location$prov_code, function(x) {
  parts <- str_split(x, "_")[[1]]
  as.integer(prov_lvls %in% parts)
}))

rm(prov_lvls)
```

Create a column with the sum of provinces per publication

```
prov %<>%
   mutate(
     prov_per_record = rowSums(
     across(
       where(is.numeric)
       )
     )
   )

prov$prov_per_record <- as.factor(prov$prov_per_record)

prov_per_record_df <- prov %>%
  group_by(prov_per_record) %>%
  summarise(n_records = length(prov_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,0))

prov_per_record_df

## # A tibble: 4 × 3
##   prov_per_record n_records percentage
##   <fct>               <int>      <dbl>
## 1 1                     517         95
## 2 2                      22          4
## 3 3                       2          0
## 4 4                       1          0
```

Create a data frame with number of publication per province

```
prov_tot <-
  prov [, -(length(prov)-1)] %>% # filter out the sum column
  summarise(across(where(is.numeric),
                   list(sum)
                   )
            )

prov_tot <- reshape2::melt(prov_tot, id.vars = NULL)

colnames(prov_tot) <- c("Province", "Count")

prov_tot %<>%
      mutate_at("Province",
                stringr::str_replace, "_1", "")
```

Ecoregions

```
ecoreg <- location
ecoreg_lvls <-
  unique(
    unlist(
      str_split(ecoreg$ecoreg_code, "\\_")
                          )
                   )

ecoreg_lvls<- ecoreg_lvls[! ecoreg_lvls %in% c("")]

ecoreg[,ecoreg_lvls] <- sapply(ecoreg_lvls,
                                      function(y){
                                        str_count(ecoreg$ecoreg_code, y)
                                        }
                                      )
```

Create a column with the number of ecoregions per study

```
ecoreg %<>%
   mutate(
     ecoreg_per_record = rowSums(
     across(
       where(is.numeric)
       )
     )
   )

ecoreg$ecoreg_per_record <- as.factor(ecoreg$ecoreg_per_record)
```

Calculate number of records by number of ecoregions in %

```r
ecoreg_per_record_df <- ecoreg %>%
  group_by(ecoreg_per_record) %>%
  summarise(n_records = length(ecoreg_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,1))

ecoreg_per_record_df
```

```
## # A tibble: 5 × 3
##   ecoreg_per_record n_records percentage
##   <fct>                 <int>      <dbl>
## 1 1                       475       87.6
## 2 2                        40        7.4
## 3 3                        18        3.3
## 4 4                         7        1.3
## 5 5                         2        0.4
```

Create a data frame with number of publication per ecoregion

```r
ecoreg_tot <-
  ecoreg [, -(length(ecoreg)-1)] %>% # filter out the sum column
  summarise(across(where(is.numeric),
                   list(sum)
                   )
            )

ecoreg_tot <- reshape2::melt(ecoreg_tot, id.vars = NULL)

colnames(ecoreg_tot) <- c("Ecoregion", "Count")

ecoreg_tot %<>%
      mutate_at("Ecoregion",
                stringr::str_replace, "_1", "")
```

Add count column to Meow and set 0 to all the ecoregions where no study was conducted

```r
colnames(ecoreg_tot)[colnames(ecoreg_tot) == 'Ecoregion'] <- 'ECO_CODE'
ecoreg_tot$ECO_CODE <- as.character(ecoreg_tot$ECO_CODE)
meow$ECO_CODE <- as.character(meow$ECO_CODE)



ecoreg_tot <-
  left_join(meow, ecoreg_tot, by = 'ECO_CODE') %>%
  mutate(Count = replace(Count, is.na(Count), 0))
```

Export count per ecoregion to csv for the map in QGIS

```r
writexl::write_xlsx(ecoreg_tot,
           here("Extracted_data_analysis","temperate","Output-Transformed_dat
a","eco_reg_tot.xlsx"))
```

Realms

```
realms <- ecoreg_tot %>%
    group_by(RLM_CODE, REALM) %>%
  summarise(Count=sum(Count))
```

```
## `summarise()` has regrouped the output.
## ℹ Summaries were computed grouped by RLM_CODE and REALM.
## ℹ Output is grouped by RLM_CODE.
## ℹ Use `summarise(.groups = "drop_last")` to silence this message.
## ℹ Use `summarise(.by = c(RLM_CODE, REALM))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
realms %<>%
  rename(Code = RLM_CODE,
         Realm = REALM)

writexl::write_xlsx(realms,
           here("Extracted_data_analysis","temperate","Output-Transformed_dat
a","Realm_count.xlsx"))
```

Export geographical distribution summary tables

```
writexl::write_xlsx(list(realms, prov_tot, prov_per_record_df, ecoreg_tot, ec
oreg_per_record_df),
           here("Extracted_data_analysis","temperate","Output-Transformed_dat
a","geographical_distribution_summary.xlsx"))
```

COUNT COLUMNS WITH COMBO

Create a subset of columns

```
count_columns <- temperate %>%
  select(File,
         meso_substrate,
         habit,
         environmentalPosition,
         observationalORexperimental,
         taxonomicTargetLevel,
         indicators,
         collectedDataType,
         molecularAnalyses)
```

Prepare data set with columns that need count only and add percentage column

```
count_columns[,] <- lapply(count_columns[,], as.factor)

count_results <- list()

for (col_name in colnames(count_columns)) {
```

```
  counts <- count(count_columns, !!sym(col_name))

  count_results[[col_name]] <- counts

  colnames(count_results[[col_name]]) <- c(col_name, "Count")

  count_results[[col_name]] %<>%
    filter(
      if_all(.cols = everything(), ~ !grepl("notapp", .x))
    ) %>% # delete rows with "notapp"
    tidyr::drop_na()

  count_results[[col_name]] %<>%
    mutate(Percentage= round(Count / sum(Count) * 100,2))
}

rm(col_name, counts)

count_results <- count_results[names(count_results) != "File"]
```

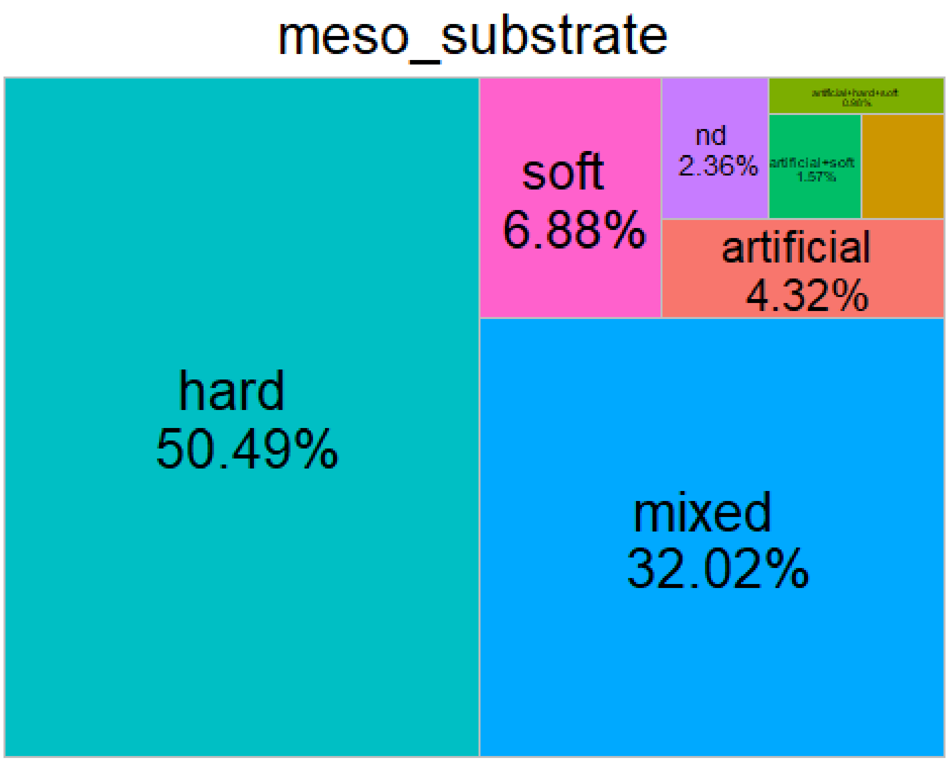
Export summary tables

```
library(openxlsx)

write.xlsx(count_results,
           file = here("Extracted_data_analysis",
                       "temperate",
                       "Output-Transformed_data",
                    "summary_counts_question_1-3.xlsx"))
```

Create treemaps charts

```
treemap_func = function(dat,var) {
  ggplot(dat,
       aes(area=Count,
           fill=.data[[var]],
           line = "black",
           label=paste0(.data[[var]],
                        " \n ",Percentage,"%"))
       )+
  treemapify::geom_treemap(layout = "squarified")+
  treemapify::geom_treemap_text(place = "centre",
                    size = 20)+
  labs(title = var)+
    theme(plot.title = element_text(hjust = 0.5, size= 20),
          legend.position = "none")

}

treemap_counts <- list()
```

```
for (col_name in colnames(count_columns[,-1])) {

  treemap_counts[[col_name]] <- treemap_func(dat= count_results[[col_name]],
var = col_name)

}

treemap_counts[[1]]
```



Save all tree maps plots

```
lapply(names(treemap_counts), function(x){
  ggsave(filename=paste0(x,".tiff"),
         plot=treemap_counts[[x]],
         units = "in", width=12, height=4, dpi=300, compression = 'lzw',
         path = here("Extracted_data_analysis",
                       "temperate",
                       "Output-Visualization",
                     "Treemap_counts"),
         create.dir = T
         )
  })

lapply(names(treemap_counts), function(x){
  ggsave(filename=paste0(x,".png"),
```

```
          plot=treemap_counts[[x]],
          units = "in", width=12, height=4, dpi=300,
          path = here("Extracted_data_analysis",
                      "temperate",
                      "Output-Visualization",
                    "Treemap_counts"),
          create.dir = T
          )
  })
```

COUNT METHODS (PURELY TAXONOMIC STUDIES EXCLUDED)

Remove purely taxonomic studies and subset columns (taxa++ was assigned to studies that led to the description of new species but where also ecological e.g. describing community for a given taxonomic group)

```
methods <- temperate %>%
  filter(taxonomic_publication == "taxa++" |  is.na(taxonomic_publication) )
 %>%
  select(File,
         meso_design,
         meso_replicates,
         temporalReplicates,
         abioticVariablesCollection,
         abioticVariables)

count_columns_method <- methods
```

Prepare data set with columns that need count only and add percentage column

```
count_columns_method[,] <- lapply(count_columns_method[,], as.factor)

count_results_method <- list()

for (col_name in colnames(count_columns_method)) {
  counts <- count(count_columns_method, !!sym(col_name))

  count_results_method[[col_name]] <- counts

  colnames(count_results_method[[col_name]]) <- c(col_name, "Count")

  count_results_method[[col_name]] %<>%
    filter(
      if_all(.cols = everything(), ~ !grepl("notapp", .x))
    ) %>%  # delete rows with "notapp"
    tidyr::drop_na()
```

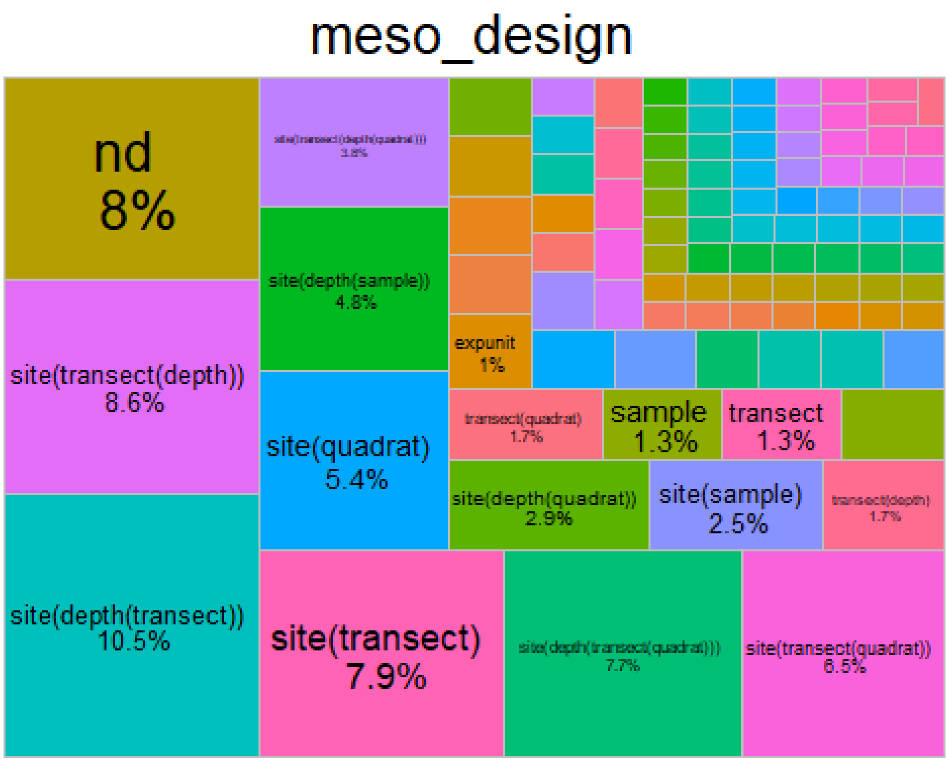
```
  count_results_method[[col_name]] %<>%
    mutate(Percentage= round(Count / sum(Count) * 100,1))
}

rm(col_name, counts)

count_results_method <- count_results_method[names(count_results_method) != "
File"]
```

Export summary tables

```
library(openxlsx)

write.xlsx(count_results_method,
           file = here("Extracted_data_analysis",                          "tem
perate",                                "Output-Transformed_data",
                    "summary_counts_question_4.xlsx"))
```

Create treemaps charts

```
treemap_func = function(dat,var) {
  ggplot(dat,
       aes(area=Count,
           fill=.data[[var]],
           line = "black",
           label=paste0(.data[[var]],
                        " \n ",Percentage,"%"))
       )+
  treemapify::geom_treemap(layout = "squarified")+
  treemapify::geom_treemap_text(place = "centre",
                    size = 20)+
  labs(title = var)+
    theme(plot.title = element_text(hjust = 0.5, size= 20),
          legend.position = "none")

}

treemap_counts <- list()

for (col_name in colnames(count_columns_method[,-1])) {

  treemap_counts[[col_name]] <- treemap_func(dat= count_results_method[[col_n
ame]], var = col_name)

}

treemap_counts[[1]]
```



Save all tree maps plots

```
lapply(names(treemap_counts), function(x){
  ggsave(filename=paste0("treemap",x,".tiff"),
         plot=treemap_counts[[x]],
         units = "in", width=12, height=4, dpi=300, compression = 'lzw',
         path = here("Extracted_data_analysis","temperate",
      "Output-Visualization","Treemaps_methods"),
         create.dir = T
         )
  })

lapply(names(treemap_counts), function(x){
  ggsave(filename=paste0("treemap",x,".png"),
         plot=treemap_counts[[x]],
         units = "in", width=12, height=4, dpi=300,
         path = here("Extracted_data_analysis",                    "tempe
rate",                        "Output-Visualization","Treemaps_methods"),
         create.dir = T
         )
  })
```

COUNT COLUMNS SPLIT

Create a list of data frame splitting values

```
split_columns <- temperate %>%
  select(meso_substrate,
         habit,
         environmentalPosition,
         observationalORexperimental,
         taxonomicTargetLevel,
         indicators,
         collectedDataType,
         molecularAnalyses)


split_columns[,] <- lapply(split_columns[,], as.factor)
```

Prepare data set with columns

```
split_results <- list()

for (col_name in colnames(split_columns)) {
  # Split the values of each column by "+"
  lvls <- unique(unlist(str_split(split_columns[[col_name]], "\\+")))

  # Initialize a list to store counts for each level
  split_counts <- sapply(lvls, function(y) {
    # Count how many times each unique level appears in the column
    str_count(split_columns[[col_name]], fixed(y))
    }
    )

  split_results[[col_name]] <- split_counts
}

## Warning in stri_count_fixed(string, pattern, opts_fixed = opts(pattern)):
empty
## search patterns are not supported

rm(lvls)
```

Add percentage column

```
split_tot <- list()

for(col_name in colnames(split_columns)) {
  split_results[[col_name]] <- as.data.frame(split_results[[col_name]])

  split_results[[col_name]] %<>% # delete rows with missing values
    select(where(function(x) any(!is.na(x)))) %>%
    select(-any_of("notapp")) %>%
    tidyr::drop_na()

    split_tot[[col_name]] <- as.data.frame(split_results[[col_name]]) %>%
```

```r
    summarise(across(where(is.numeric), list(sum)))

  split_tot[[col_name]]  <- reshape2::melt(split_tot[[col_name]], id.vars = N
ULL)

  colnames(split_tot[[col_name]]) <- c(col_name, "Count")

  split_tot[[col_name]] %<>%
        mutate_at(col_name,
                  stringr::str_replace, "_1", "")


  split_tot[[col_name]] %<>%
      mutate(Percentage= round(Count / nrow(split_results[[col_name]] ) * 100,
2))

}

rm(col_name)
```

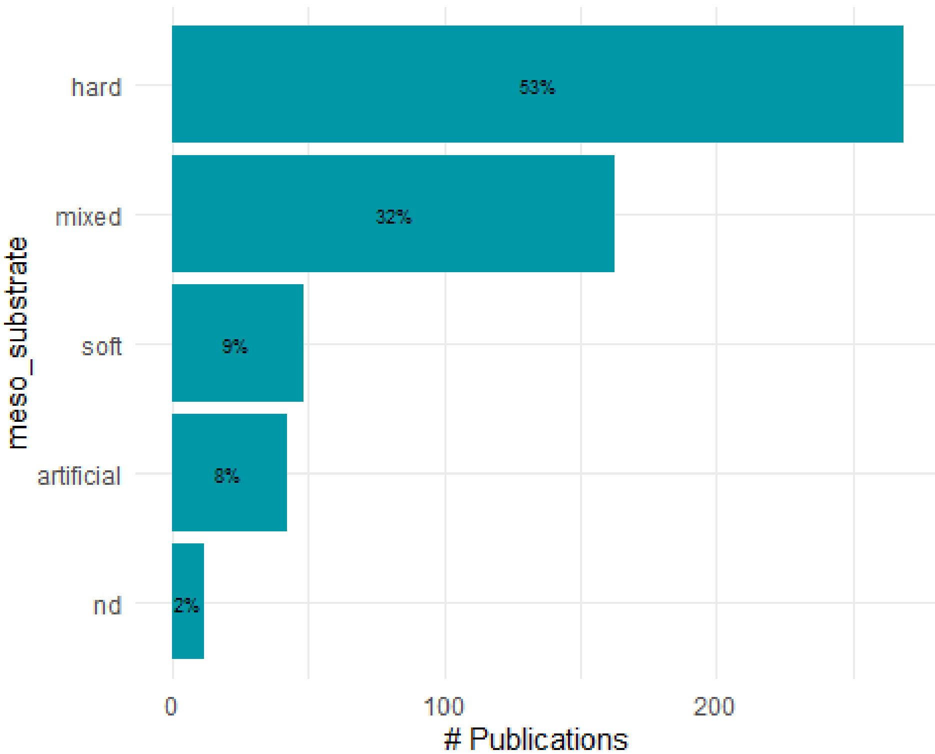
Export summary tables

```r
library(openxlsx)

write.xlsx(split_tot,
           file = here("Extracted_data_analysis",                       "tem
perate",                     "Output-Transformed_data",
                    "summary_split_counts_question_1_3.xlsx"))
```

Create bar charts for each collected parameter

```r
barchart_func = function(dat,var) {

 ggplot(dat,
       aes(x = reorder(.data[[var]], Count), y = Count, fill= .data[[var]]))
+
  geom_bar(stat = "identity", fill = "#0097A7") +
  geom_text(
    data = dat %>% filter(Percentage >= 2),
    aes(label = paste0(round(Percentage, 0), "%")),
    position = position_stack(vjust =0.5),
    color = "black",
    size = 2.5
  )+
  labs(y = "# Publications", x = var) +
  theme_minimal() +
  coord_flip()+  # optional: flips for horizontal bars
  theme(legend.position ="none")
```

```
}

barchart_counts <- list()

for (col_name in colnames(split_columns)) {

  barchart_counts[[col_name]] <- barchart_func(dat= split_tot[[col_name]], va
r = as.character(col_name))

}

barchart_counts[[1]]
```




```
lapply(names(barchart_counts), function(x){
  ggsave(filename=paste0("bar_",x,".tiff"),
         plot=barchart_counts[[x]],
         units = "in", width=12, height=4, dpi=300, compression = 'lzw',
         path = here("Extracted_data_analysis",                        "tempe
rate",                        "Output-Visualization","Barcharts_split"),
         create.dir = T
         )
  })

## [[1]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
```

```
ut-Visualization/Barcharts_split/bar_meso_substrate.tiff"
##
## [[2]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_habit.tiff"
##
## [[3]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_environmentalPosition.tiff"
##
## [[4]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_observationalORexperimental.tiff"
##
## [[5]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_taxonomicTargetLevel.tiff"
##
## [[6]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_indicators.tiff"
##
## [[7]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_collectedDataType.tiff"
##
## [[8]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_molecularAnalyses.tiff"
```

```
lapply(names(barchart_counts), function(x){
  ggsave(filename=paste0("bar_",x,".png"),
         plot=barchart_counts[[x]],
         units = "in", width=12, height=4, dpi=300,
         path = here("Extracted_data_analysis",                        "tempe
rate",                        "Output-Visualization","Barcharts_split"),
         create.dir = T
         )
  })
```

```
## [[1]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
```

```
ut-Visualization/Barcharts_split/bar_meso_substrate.png"
##
## [[2]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_habit.png"
##
## [[3]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_environmentalPosition.png"
##
## [[4]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_observationalORexperimental.png"
##
## [[5]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_taxonomicTargetLevel.png"
##
## [[6]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_indicators.png"
##
## [[7]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_collectedDataType.png"
##
## [[8]]
## [1] "C:/Users/claud/Documents/COMUNE/STUDIO_LAVORO/Data_analysis/Projects/
Mesophotic_systematic_lit_rev_analysis/Extracted_data_analysis/temperate/Outp
ut-Visualization/Barcharts_split/bar_molecularAnalyses.png"
```

COUNT METHODS (PURELY TAXONOMIC STUDIES EXCLUDED)

Create data frame with subset

```
split_columns_methods <- methods

split_columns_methods[,] <- lapply(split_columns_methods[,], as.factor)
```

Prepare data set with columns

```r
split_results_methods <- list()

for (col_name in colnames(split_columns_methods)) {
  # Split the values of each column by "+"
  lvls <- unique(unlist(str_split(split_columns_methods[[col_name]], "\\+")))

  # Initialize a list to store counts for each level
  split_counts_methods <- sapply(lvls, function(y) {
    # Count how many times each unique level appears in the column
    str_count(split_columns_methods[[col_name]], fixed(y))
    }
    )

  split_results_methods[[col_name]] <- split_counts_methods
}

rm(lvls)
```

Add percentage column

```r
split_tot_methods <- list()

for(col_name in colnames(split_columns_methods)) {
  split_results_methods[[col_name]] <- as.data.frame(split_results_methods[[c
ol_name]])

  split_results_methods[[col_name]] %<>% # delete rows with missing values
    select(where(function(x) any(!is.na(x)))) %>%
    select(-any_of("notapp")) %>%
    tidyr::drop_na()

    split_tot_methods[[col_name]] <- as.data.frame(split_results_methods[[col
_name]]) %>%
    summarise(across(where(is.numeric), list(sum)))

  split_tot_methods[[col_name]]  <- reshape2::melt(split_tot_methods[[col_nam
e]], id.vars = NULL)

  colnames(split_tot_methods[[col_name]]) <- c(col_name, "Count")

  split_tot_methods[[col_name]] %<>%
        mutate_at(col_name,
                  stringr::str_replace, "_1", "")


  split_tot_methods[[col_name]] %<>%
      mutate(Percentage= round(Count / nrow(split_results_methods[[col_name]]
 ) * 100,2))
```

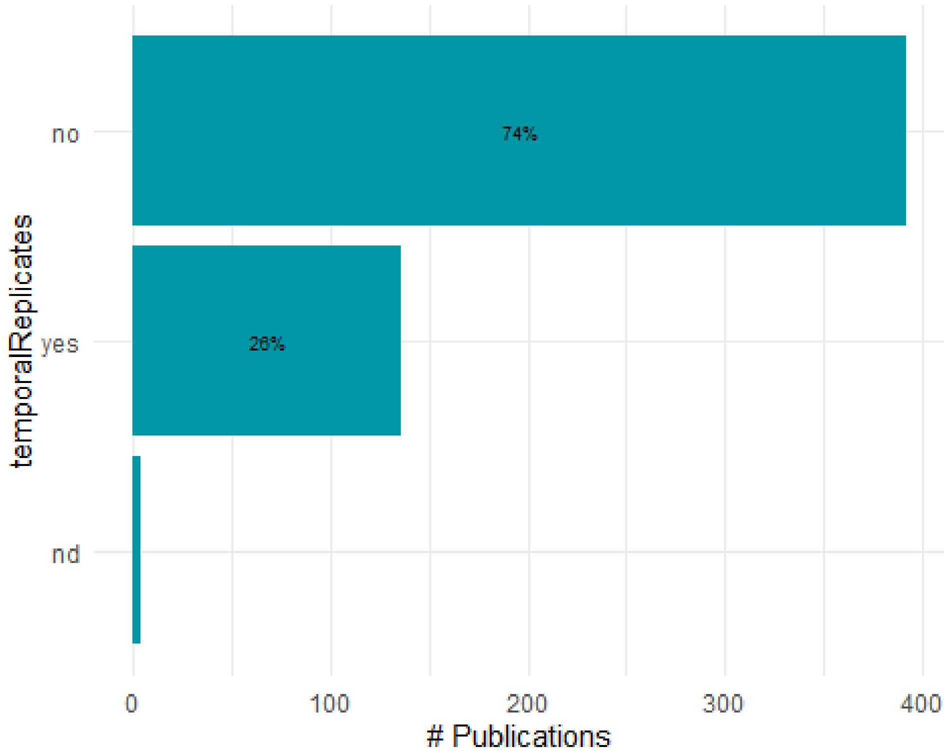
```
}

rm(col_name)
```

Export summary tables

```
library(openxlsx)

write.xlsx(split_results_methods,
           file = here("Extracted_data_analysis",                        "tem
perate",                          "Output-Transformed_data",
                     "summary_split_counts_question_4.xlsx"))
```

Create bar charts per each parameter collected

```
barchart_func = function(dat,var) {

 ggplot(dat,
       aes(x = reorder(.data[[var]], Count), y = Count, fill= .data[[var]]))
+
  geom_bar(stat = "identity", fill = "#0097A7") +
  geom_text(
    data = dat %>% filter(Percentage >= 2),
    aes(label = paste0(round(Percentage, 0), "%")),
    position = position_stack(vjust =0.5),
    color = "black",
    size = 2.5
  )+
  labs(y = "# Publications", x = var) +
  theme_minimal() +
  coord_flip()+  # optional: flips for horizontal bars
  theme(legend.position ="none")

}

barchart_split_methods <- list()

for (col_name in colnames(split_columns_methods)) {

  barchart_split_methods[[col_name]] <- barchart_func(dat= split_tot_methods
[[col_name]], var = as.character(col_name))

}

barchart_split_methods[[4]]
```



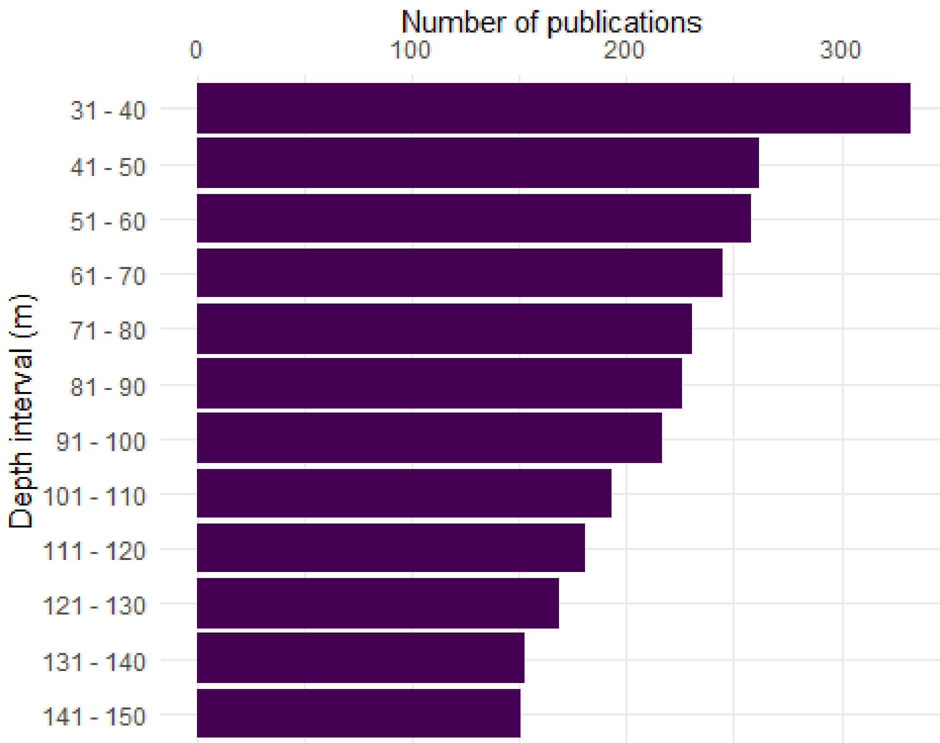
Export bar charts in a dedicated folder

```
lapply(names(barchart_counts), function(x){
  ggsave(filename=paste0("bar_",x,".tiff"),
         plot=barchart_counts[[x]],
         units = "in", width=12, height=4, dpi=300, compression = 'lzw',
         path = here("Extracted_data_analysis",                          "tempe
rate",                         "Output-Visualization","Barcharts_split_methods
"), create.dir = T
         )
  })

lapply(names(barchart_counts), function(x){
  ggsave(filename=paste0("bar_",x,".png"),
         plot=barchart_counts[[x]],
         units = "in", width=12, height=4, dpi=300,
         path = here("Extracted_data_analysis",                          "tempe
rate",                         "Output-Visualization","Barcharts_split"), crea
te.dir = T
         )
  })
```

PUBLICATIONS PER DEPTH

```
temperate$mininumSurveyedDepthInMeters <- as.numeric(temperate$mininumSurveye
dDepthInMeters)
```

```
## Warning: NAs introduced by coercion

temperate$maximumSurveyedDepthInMeters <- as.numeric(temperate$maximumSurveye
dDepthInMeters)

## Warning: NAs introduced by coercion

# Define depth bins
bins <- seq(31, 151, by = 10)

# Create a data frame to hold bin counts
pubs_depth <- data.frame(
  bin_start = bins[-length(bins)],
  bin_end = bins[-1]-1,
  count = 0
)

# For each bin, count how many studies cover that depth range
for (i in 1:nrow(pubs_depth)) {
  bin_start <- pubs_depth$bin_start[i]
  bin_end <- pubs_depth$bin_end[i]

  pubs_depth$count[i] <- sum(temperate$maximumSurveyedDepthInMeters >= bin_st
art & temperate$mininumSurveyedDepthInMeters < bin_end, na.rm = TRUE)
}

pubs_depth$depth_bin <- paste(pubs_depth$bin_start, "-", pubs_depth$bin_end)

pubs_depth$depth_bin <- reorder(pubs_depth$depth_bin, -pubs_depth$bin_start)


ggplot(pubs_depth, aes(x = factor(depth_bin), y = count)) +
  geom_bar(stat = "identity", fill = "#440154") +
  labs(x = "Depth interval (m)", y = "Number of publications") +
  theme_minimal()+
  coord_flip()+
  scale_y_continuous(position="right")
```



Export pubs per depth bar plot

PLATFORM BY DEPTH

Create a platform by depth data frame and keep meso_samplingPlatform

```
platform_depth <- temperate %>%
  select(File,
         mininumSurveyedDepthInMeters,
         maximumSurveyedDepthInMeters,
         meso_direct_indirect,
         meso_samplingPlatform,
         max_depth_hov, min_depth_hov,
         max_depth_rov, min_depth_rov,
         max_depth_ccr, min_depth_ccr,
         max_depth_open, min_depth_open,
         max_depth_scuba, min_depth_scuba
         )
```

Split the platform column in one binary column per platform

```
platform_lvls <-
  unique(
    unlist(
      str_split(platform_depth$meso_samplingPlatform, "\\+")
                                )
```

```
                        )

platform_lvls<- platform_lvls[! platform_lvls %in% c("",NA,"notapp",",","nd")]

platform_depth[,platform_lvls] <- sapply(platform_lvls,
                                      function(y){
                                        str_count(platform_depth$meso_samplin
gPlatform, y)
                                        }
                                      )

rm(platform_lvls)

platform_depth[,-c(1:15)] <- lapply(platform_depth[,-c(1:15)], as.numeric)

platform_depth[,c(1:15)] <- lapply(platform_depth[,c(1:15)], as.character)


platform_depth %<>%
  rename(open_direct = open,
         ccr_direct = ccr,
         scuba_direct = scuba)  %>%
  select(-meso_samplingPlatform)
```

Create a summary table with # studies per platform

```
platform_summary <- platform_depth %>%
  summarise(across(where(is.numeric), ~ sum(.x, na.rm = TRUE)))

platform_summary <- as.data.frame(t(platform_summary))

platform_summary %<>%
  mutate(Platform = rownames(platform_summary)) %>%
  rename(Count = V1) %>%
  relocate(Count,.after = where(is.character))

platform_summary

##                  Platform Count
## rov                   rov   276
## trawl               trawl     9
## acou                 acou    97
## dredge             dredge    16
## scuba_direct scuba_direct    88
## hov                   hov    38
## open_direct   open_direct   133
## grab                 grab    39
## drop                 drop    22
## fishery           fishery    35
```

```
## tcs                          tcs    24
## ccr_direct            ccr_direct    12
## auv                          auv    20
## bruv                        bruv     7
## drift                      drift     2
## bdrop                      bdrop     1
## station                  station     1
## trap                        trap     2
```

Create a column with the number of platforms per study

```
platform_per_record <- platform_depth %>%
  filter(meso_direct_indirect != "nd") %>%
   mutate(
     platform_per_record = rowSums(
     across(
       where(is.numeric)
       )
     )
   )

platform_per_record$platform_per_record <- as.factor(platform_per_record$plat
form_per_record)
```

Calculate number of records by number of platforms in %

```
platform_per_record <- platform_per_record %>%
  group_by(platform_per_record) %>%
  summarise(n_records = length(platform_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,1))

platform_per_record

## # A tibble: 5 × 3
##   platform_per_record n_records percentage
##   <fct>                   <int>      <dbl>
## 1 1                         339       62.8
## 2 2                         136       25.2
## 3 3                          54       10
## 4 4                           6        1.1
## 5 5                           5        0.9
```

Keep most common visual platforms only

```
platform_depth_mod <- cbind(platform_depth[,c(1:14)],
                            platform_depth$ccr_direct,
                            platform_depth$open_direct,
                            platform_depth$scuba_direct,
                            platform_depth$hov,
                            platform_depth$rov)
```

```
names(platform_depth_mod) <- gsub("platform_depth\\$", "", names(platform_dep
th_mod))
```

Set max depth of studies with max depth > 150 to 150 so they fit into the plot

```
platform_depth_long <- within(platform_depth_mod, rm(mininumSurveyedDepthInMe
ters, maximumSurveyedDepthInMeters, max_depth_direct, meso_direct_indirect, h
ov, rov, open_direct, ccr_direct, scuba_direct))

platform_depth_long[,-1] <- lapply(platform_depth_long[,-1], as.numeric)

str(platform_depth_long)

## 'data.frame':    542 obs. of  11 variables:
##  $ File           : chr  "AchilleosEtAl_2020_Bryozoan_diversity_Cyprus.md"
 "AdamsEtAl_2020_Rhodolith_bed_discovered.md" "AdamsEtAl_2023_Patterns_potent
ial_drivers.md" "Aguado-GimenezRuiz-Fernandez_2012_Influence_experimental_fis
h.md" ...
##  $ max_depth_hov  : num  NA NA NA NA NA 100 NA NA NA NA ...
##  $ min_depth_hov  : num  NA NA NA NA NA 0 NA NA NA NA ...
##  $ max_depth_rov  : num  477 220 100 NA 707 NA 110 70 133 125 ...
##  $ min_depth_rov  : num  118 30 30 NA 100 NA 20 12 106 48 ...
##  $ max_depth_ccr  : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ min_depth_ccr  : num  NA NA NA NA NA NA NA NA NA NA ...
##  $ max_depth_open : num  NA NA NA NA NA NA 45 NA NA NA ...
##  $ min_depth_open : num  NA NA NA NA NA NA 10 NA NA NA ...
##  $ max_depth_scuba: num  NA NA NA 40 NA NA NA NA NA NA ...
##  $ min_depth_scuba: num  NA NA NA 30 NA NA NA NA NA NA ...

platform_depth_long  %<>%
  mutate(max_depth_scuba = case_when(max_depth_scuba >= 150 ~ 150,
                               TRUE ~ max_depth_scuba)
       ) %>%
  mutate(max_depth_hov = case_when(max_depth_hov >= 150 ~ 150,
                               TRUE ~ max_depth_hov)
       ) %>%
  mutate(max_depth_rov = case_when(max_depth_rov >= 150 ~ 150,
                               TRUE ~ max_depth_rov)
       ) %>%
  mutate(max_depth_ccr = case_when(max_depth_ccr >= 150 ~ 150,
                               TRUE ~ max_depth_ccr)) %>%
  mutate(max_depth_open = case_when(max_depth_open >= 150 ~ 150,
                               TRUE ~ max_depth_open))
```

Set min depth of studies with min depth <30 to 30 so they fit into the plot

```
platform_depth_long %<>%
  mutate(min_depth_scuba = case_when(min_depth_scuba <= 30 ~ 30,
                               TRUE ~ min_depth_scuba)
       ) %>%
  mutate(min_depth_hov = case_when(min_depth_hov <= 30 ~ 30,
```

```r
                                    TRUE ~ min_depth_hov)
        ) %>%
  mutate(min_depth_rov = case_when(min_depth_rov <= 30 ~ 30,
                                    TRUE ~ min_depth_rov)
        ) %>%
  mutate(min_depth_ccr = case_when(min_depth_ccr <= 30 ~ 30,
                                    TRUE ~ min_depth_ccr)) %>%
  mutate(min_depth_open = case_when(min_depth_open <= 30 ~ 30,
                                    TRUE ~ min_depth_open))
```

Convert platform_depth_long to long format

```r
platform_min_depth_long <- platform_depth_long %>%
  select(-c(max_depth_hov, max_depth_rov, max_depth_open, max_depth_scuba, ma
x_depth_ccr))



platform_min_depth_long[,c(1,2)] <- lapply(platform_min_depth_long[,c(1,2)],
as.factor)

platform_min_depth_long %<>%
  reshape2::melt(platform_min_depth_long,
                 id.vars = c("File"),
                 measure.vars = grep("^min_depth_|^max_depth_",
                                     names(platform_min_depth_long),
                                     value = TRUE),
                 variable.name = "Platform",
                 value.name = "min_depth"
                 )

## Warning: attributes are not identical across measure variables; they will
be
## dropped

platform_max_depth_long <- platform_depth_long %>%
    select(-c(min_depth_hov, min_depth_rov, min_depth_open, min_depth_scuba,
min_depth_ccr))

platform_max_depth_long %<>%
    reshape2::melt(platform_max_depth_long,
                 id.vars = c("File"),
                 measure.vars = grep("^min_depth_|^max_depth_",
                                     names(platform_max_depth_long),
                                     value = TRUE),
                 variable.name = "Platform",
                 value.name = "max_depth"
                 )

platform_depth_lat_long <- cbind(platform_min_depth_long, platform_max_depth_
long$max_depth)
```

```r
rm(platform_min_depth_long, platform_max_depth_long)

platform_depth_lat_long %<>%
  rename(max_depth =`platform_max_depth_long$max_depth`)

platform_depth_lat_long$Platform <- gsub("min_depth_", "", platform_depth_lat
_long$Platform)

platform_depth_lat_long$Platform <- gsub("max_depth_", "", platform_depth_lat
_long$Platform)
```

Set proper labelling for platform

```r
platform_depth_lat_long %<>%
  mutate(Platform = case_when(Platform == "ccr" ~ "CCR",
                              Platform == "scuba" ~ "SCUBA",
                              Platform == "open" ~ "OC",
                              Platform == "rov" ~ "ROV",
                              Platform == "hov" ~ "HOV",
                              TRUE ~ Platform)
         ) %>%
  mutate(Platform = as.factor(Platform))
```

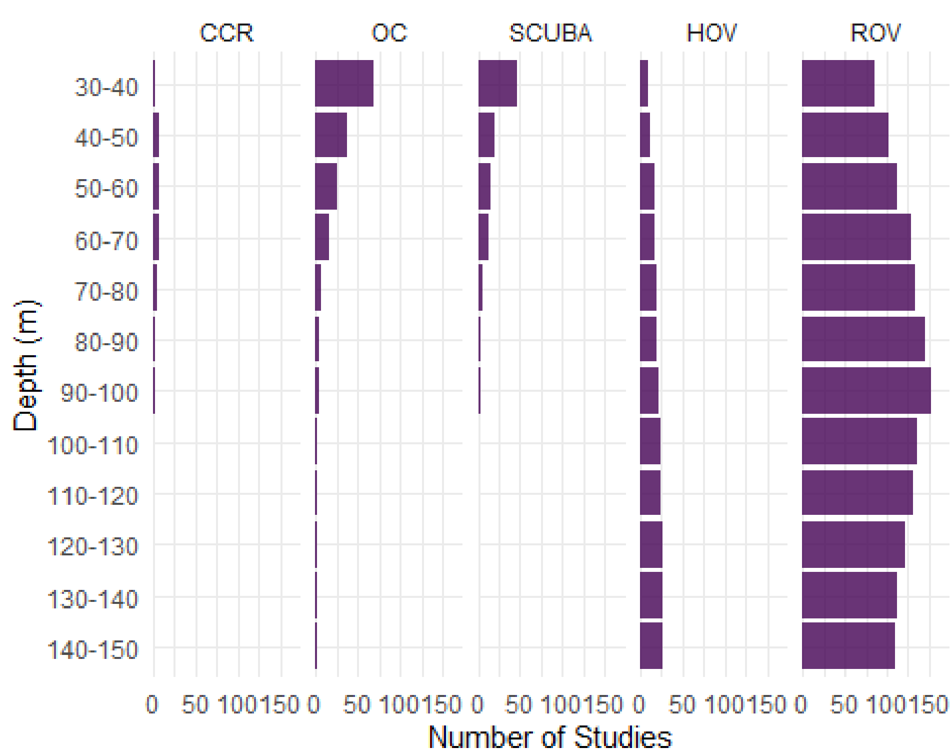
Add visual column to reorder platform

```r
platform_depth_lat_long %<>%
  mutate(Visual = as.factor(case_when(Platform == "OC" |
                                      Platform == "SCUBA" |
                                      Platform == "CCR" ~ "direct",
                            TRUE ~ "indirect" )
  )
  )

platform_depth_lat_long %<>%
  mutate(Platform = forcats::fct_reorder(Platform, as.integer(Visual)))
```

Create a bar plot with depth bins by platform and latitudinalRegion

```r
depth_bins <- tibble(
  bin_start = seq(-30, -140, by = -10),
  bin_end   = seq(-40, -150, by = -10)
) %>%
  mutate(bin_label = paste0(abs(bin_start), "-", abs(bin_end)))

platform_depth_lat_long <- platform_depth_lat_long %>%
  mutate(
    min_depth = as.numeric(min_depth),  # positive e.g., 30
    max_depth = as.numeric(max_depth)   # positive e.g., 150
  )
```

```
expanded_data <- platform_depth_lat_long %>%
  tidyr::crossing(depth_bins) %>%
  # Convert study range to negative depths: [-max_depth, -min_depth]
  filter(bin_end >= -max_depth, bin_start <= -min_depth) %>%
  mutate(
    bin_label = factor(bin_label, levels = rev(depth_bins$bin_label))
  )

# Plot with count labels on the right
platform_depth_plot <- ggplot(expanded_data, aes(x = bin_label)) +
  geom_bar(position = "stack", alpha= 0.8, fill= "#440154") +
  scale_y_continuous(name = "Number of Studies",
                     expand = expansion(mult = c(0, 0.15))) +  # Make room fo
r text
  scale_x_discrete(name = "Depth (m)") +
  coord_flip() +  # Depth on y-axis (shallow at top)
  facet_wrap(~ Platform, nrow = 1) +
  theme_minimal() +
  theme(legend.position = "none")

platform_depth_plot
```



Export bar plot of most common platform per depth interval

```
ggsave("platform_depth_plot.tiff",
       plot = platform_depth_plot,
       units = "in", width = 8, height = 6, dpi = 300, compression = 'lzw',
       path = here ("Extracted_data_analysis",                         "temper
ate",                             "Output-Visualization"))

ggsave("platform_depth_plot.png",
       plot = platform_depth_plot,
       units = "in", width = 8, height = 5,
       path = here ("Extracted_data_analysis", "temperate",
    "Output-Visualization"))
```

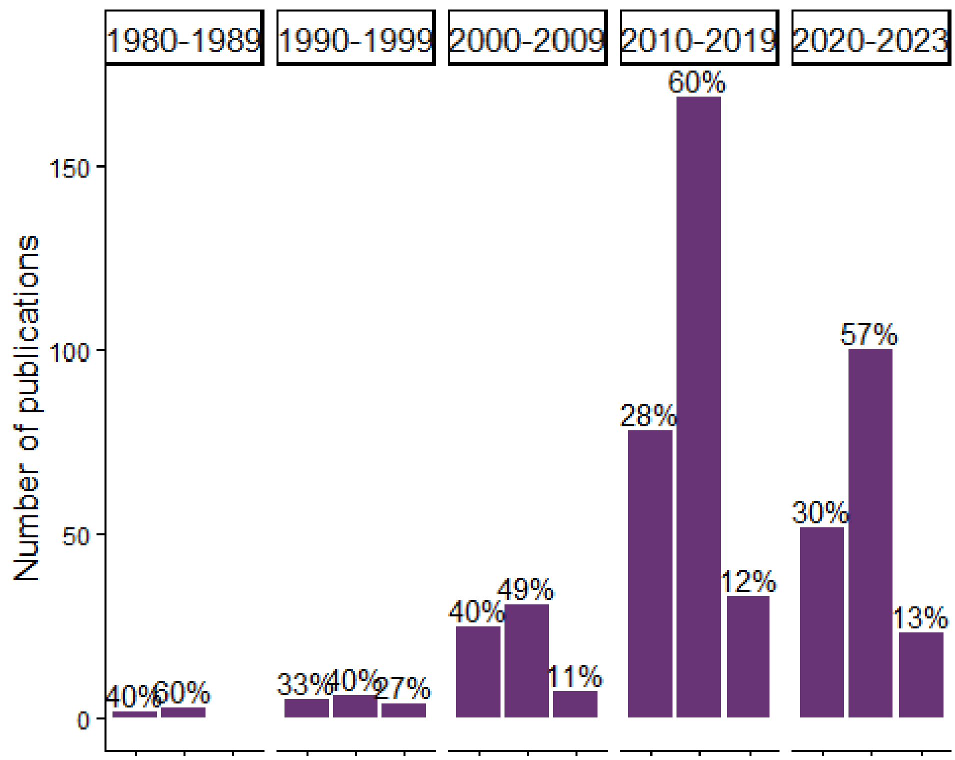
DIRECT - INDIRECT BY DECADE

```
dir_ind_by_decade <- temperate

dir_ind_by_decade %<>%
         mutate(decade = case_when(publicationYear >= 1960 & publicationYear
< 1970 ~ "1970-1979",
                                 publicationYear >= 1970 & publicationYear <
1980 ~ "1970-1979",
                                 publicationYear >= 1980 & publicationYear <
1990 ~ "1980-1989",
                                 publicationYear >= 1990 & publicationYear <
2000 ~ "1990-1999",
                                 publicationYear >= 2000 & publicationYear <
2010 ~ "2000-2009",
                                 publicationYear >= 2010 & publicationYear <
2020 ~ "2010-2019",
                                 publicationYear >= 2020 ~ "2020-2023")
              )

dir_ind_by_decade_lat <- dir_ind_by_decade

dir_ind_by_decade <- dir_ind_by_decade [, c("meso_direct_indirect","decade")]
 %>%
  group_by(decade,meso_direct_indirect) %>%
  tally()

colnames(dir_ind_by_decade) <- c("Decade", "Technique","Count")

unique(dir_ind_by_decade$Technique)

## [1] "direct"   "indirect" "mixed"    "nd"

dir_ind_by_decade %<>%
  filter(Technique != "nd" & Technique != "NA") %>% #remove nd from viz becau
se they accounted for a really small percentage
```

```
  group_by(Decade) %>%
  mutate(Percentage = round(Count / sum(Count) * 100,1))
```

Make bar plot of technique by decade

```
technique_decade_barplot <- ggplot(subset(dir_ind_by_decade,
              dir_ind_by_decade$Decade != "1970-1979"),
       aes(x=Technique, y=Count, label = Technique)
       ) +
  geom_bar(position="dodge",
           stat="identity", alpha= 0.8, fill = "#440154" ) +
  geom_text(aes(label = paste(round(Percentage,0),"%", sep="")),
            parse = F,
            vjust = -0.25,
            size=3.9)+
  facet_grid(~ Decade,
             switch = "y") +
  theme_classic() +
  theme(legend.position = "bottom",
        legend.text = element_text(size = 10),
        axis.text.x = element_blank(),
        axis.text.y = element_text(size=8.7),
        axis.title.x = element_blank(),
        axis.title.y = element_text(size=13),
        strip.text = element_text(size = 13))+
  labs(y= "Number of publications")

technique_decade_barplot
```



Export bar plot of technique by decade

```
ggsave("technique_decade_barplot.tiff",
       plot = technique_decade_barplot,
       units = "in", width = 7, height = 6, dpi = 300, compression = 'lzw',
       path = here ("Extracted_data_analysis",                        "temper
ate",                             "Output-Visualization"))

ggsave("technique_decade_barplot.png",
       plot = technique_decade_barplot,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                        "temper
ate",                             "Output-Visualization"))
```

FIELD & INDICATORS

Create subset for field & indicators

```
field_merged <- temperate[,c(59:length(temperate))]

field_merged[,] <- lapply(field_merged[,], as.numeric)

colSums(field_merged)

##                    field_cons                  field_commdesc
##                            97                             257
```

```
##                   field_map                  field_impass
##                         125                           187
##                 field_troph                    field_taxo
##                          28                            42
##                  field_dist                    field_aqua
##                         154                             3
##                field_method                  field_physio
##                          51                            40
##                  field_func                 field_habtype
##                          30                             2
##               field_fishery                   field_stock
##                          45                            50
##                 field_paleo                   field_phylo
##                           1                            21
##                  field_rest                 field_archaeo
##                           8                             1
##                 field_micro             field_biomedicine
##                           3                             0
##                  field_chem        field_Molecular_ecology
##                           0                            41
##             field_Acoustics field_environmental_drivers
##                          97                           124
##             field_Behaviour           field_Geomorphology
##                          72                            43
##              field_Taxonomy
##                          35
```

Select only the most common fields

```r
field_merged <- field_merged[, colSums(field_merged) > 50]

field_merged$field_Taxonomy <- NULL

colSums(field_merged)
```

```
##                  field_cons                field_commdesc
##                          97                           257
##                   field_map                  field_impass
##                         125                           187
##                  field_dist                  field_method
##                         154                            51
##             field_Acoustics field_environmental_drivers
##                          97                           124
##             field_Behaviour
##                          72
```

```r
field_merged <- field_merged[, order(colSums(field_merged), decreasing = F)]

colSums(field_merged)
```

```
##                field_method             field_Behaviour
##                          51                          72
##                  field_cons             field_Acoustics
##                          97                          97
## field_environmental_drivers                   field_map
##                         124                         125
##                  field_dist                field_impass
##                         154                         187
##              field_commdesc
##                         257
```

Rename fields with long version names

```
field_merged %<>%
  rename(`Biodiversity and \n community structure (256)` = field_commdesc,
         `Distribution and \n connectivity (154)` = field_dist,
         `Disturbances (187)` = field_impass,
         `Conservation and \n management (97)` = field_cons,
         `Mapping (125)` = field_map,
         `Methods (51)` = field_method ,
         `Behaviour (71)` = field_Behaviour ,
         `Acoustics (97)` = field_Acoustics ,
         `Environmental drivers and \n spatio-temporal patterns (123)` = fiel
d_environmental_drivers
         )
colSums(field_merged)
```

```
##                                                Methods (51)
##                                                          51
##                                              Behaviour (71)
##                                                          72
##                         Conservation and \n management (97)
##                                                          97
##                                              Acoustics (97)
##                                                          97
## Environmental drivers and \n spatio-temporal patterns (123)
##                                                         124
##                                               Mapping (125)
##                                                         125
##                      Distribution and \n connectivity (154)
##                                                         154
##                                          Disturbances (187)
##                                                         187
##               Biodiversity and \n community structure (256)
##                                                         257
```

Prepare the indicator matrix

```
ind_matrix <- split_results[["indicators"]]
```

Select only the most common indicators

```r
colSums(ind_matrix)
```

```
##        Abundance         Diversity           Habitat     Morphological 
##              245               265               118               100 
##             taxo           Biomass             Cover            impact 
##               32                40               173                11 
##             surv           Density            growth            Health 
##               13               174                19                62 
##    Physiological              Size            ALDFGs            Litter 
##               35               149                61                50 
##              NIS         Mortality       Recruitment              func 
##               18                13                10                10 
##          Trophic geneticSequences         Behaviour               sex 
##               25                42                20                15 
##    economicValue        complexity            ecoser       orientation 
##                1                 3                 3                 1 
##          disease 
##                1
```

```r
ind_common_matrix <- ind_matrix[, colSums(ind_matrix) > 30]

colSums(ind_common_matrix)
```

```
##        Abundance         Diversity           Habitat     Morphological 
##              245               265               118               100 
##             taxo           Biomass             Cover           Density 
##               32                40               173               174 
##           Health     Physiological              Size            ALDFGs 
##               62                35               149                61 
##           Litter geneticSequences 
##               50                42
```

```r
ind_common_matrix <- ind_common_matrix[, order(colSums(ind_common_matrix), de
creasing = TRUE)]

ind_common_matrix <- as.data.frame(ind_common_matrix)

ind_common_matrix <- ind_common_matrix[!purrr::map_lgl(ind_common_matrix, ~ a
ll(is.na(.)))] # remove empty column
```

Remove taxonomy studies which have a dedicated section

```r
ind_common_matrix$taxo <- NULL
field_merged$field_taxo <- NULL

colSums(ind_common_matrix)
```

```
##        Diversity         Abundance           Density             Cover 
##              265               245               174               173 
##             Size           Habitat     Morphological            Health 
##              149               118               100                62
```

```
##          ALDFGs           Litter geneticSequences          Biomass
##              61               50               42               40
##    Physiological
##              35
```

Rename indicators with long version names

```
ind_common_matrix %<>%
  rename(`Abundance \n (245)` = Abundance,
         `Cover \n (171)` = Cover,
         `Diversity \n (264)` = Diversity,
         `Habitat \n (118)` = Habitat,
         `Morphology \n (99)` = Morphological,
         `Size \n (149) ` = Size,
         `Health \n (62) ` = Health,
         `Density \n (175)` = Density,
         `Physiological \n indicator \n (35)` = Physiological,
         `ALDFGs  \n (61)` = ALDFGs ,
         `Litter \n (50)` = Litter,
         `Biomass \n (40)` = Biomass,
         `Genetic \n sequence \n (41)` = geneticSequences
         )

colSums(ind_common_matrix)
```

```
##                Diversity \n (264)                Abundance \n (245)
##                               265                               245
##                  Density \n (175)                    Cover \n (171)
##                               174                               173
##                    Size \n (149)                   Habitat \n (118)
##                               149                               118
##                Morphology \n (99)                   Health \n (62)
##                               100                                62
##                   ALDFGs  \n (61)                    Litter \n (50)
##                                61                                50
##          Genetic \n sequence \n (41)                 Biomass \n (40)
##                                42                                40
## Physiological \n indicator \n (35)
##                                35
```

Multiply the transpose of the field matrix with the indicator matrix

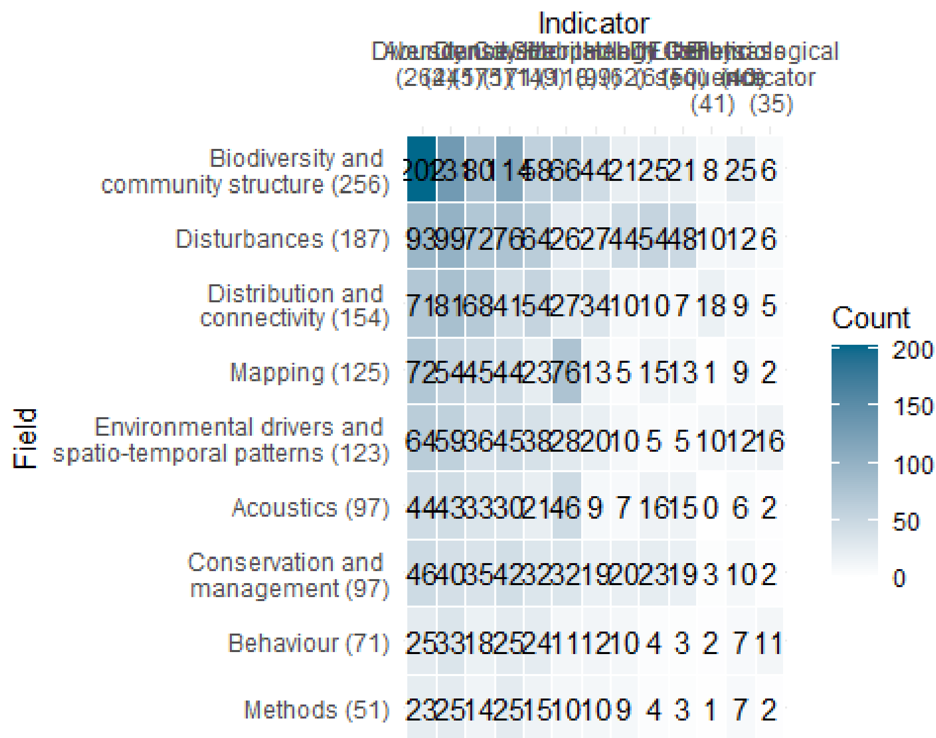
```
co_occurrence_matrix <- t(as.matrix(field_merged)) %*% as.matrix(ind_common_m
atrix)
```

Reshape the co-occurrence matrix for plotting}

```
co_occurrence_df <- reshape2::melt(co_occurrence_matrix)

colnames(co_occurrence_df) <- c("Field", "Indicator", "Count")
```

Plot the heatmap

```
field_ind_heat <- ggplot(co_occurrence_df, aes(x = Indicator, y = Field, fill
 = Count)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "white", high = "deepskyblue4") +
  geom_text(aes(label = Count), color = "black") +
  theme_minimal()+
  scale_x_discrete(position="top")

field_ind_heat
```



Export plot

```
ggsave("field_ind_heat.png",
       units = "in", width = 12.5, height = 5.5,
       path = here ("Extracted_data_analysis", "temperate",
      "Output-Visualization"))
```

Calculate # of ecological indicators per record

Create a column with the sum of ecological indicators values to see how many different sampling units were adopted by each study

```
ind_summary <- as.data.frame(split_results[["indicators"]])

ind_summary <- ind_summary[!purrr::map_lgl(ind_summary, ~ all(is.na(.)))] # r
```

```
emove empty column

ind_summary %<>%  rowwise() %>%
   mutate(ind_per_record =  sum(c_across(where(is.numeric))))
```

Calculate number of records by number of sampling units in %

```
ind_summary$ind_per_record <- as.factor(ind_summary$ind_per_record)

ind_per_record <- ind_summary %>%
  group_by(ind_per_record) %>%
  summarise(n_records = length(ind_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,1))

ind_per_record

## # A tibble: 9 × 3
##   ind_per_record n_records percentage
##   <fct>              <int>      <dbl>
## 1 1                     58       10.7
## 2 2                    153       28.2
## 3 3                    135       24.9
## 4 4                    108       19.9
## 5 5                     49        9
## 6 6                     16        3
## 7 7                     18        3.3
## 8 8                      4        0.7
## 9 9                      1        0.2
```

SAMPLING UNIT

Prepare a data frame for sampling unit

```
sampling_df <- temperate %>%
  filter(is.na(taxonomic_publication)) %>% # remove purely taxonomic studies
  select(File,recordTitle,DOI,publicationYear,mininumSurveyedDepthInMeters,ma
ximumSurveyedDepthInMeters,samplingUnit)

sampling_df %>% tidyr::drop_na(samplingUnit)
sampling_df[!is.na(sampling_df$samplingUnit),]

sampling_df %<>%
  filter(samplingUnit != "notapp")

sampling_df$samplingUnit <- as.character(sampling_df$samplingUnit)
```

Prepare columns per sampling units

```
sampling_lvls <- unique(unlist(str_split(sampling_df$samplingUnit,
                                         "\\+")
                               )
                        )

rownames(is.na(sampling_df$samplingUnit))

## NULL

sampling_df[,sampling_lvls] <- sapply(sampling_lvls,
                                      function(y){
                                        str_count(sampling_df$samplingUnit, y)
                                        }
                                      )

rm(sampling_lvls)

colnames(sampling_df)

##  [1] "File"                            "recordTitle"
##  [3] "DOI"                             "publicationYear"
##  [5] "mininumSurveyedDepthInMeters"    "maximumSurveyedDepthInMeters"
##  [7] "samplingUnit"                    "sample"
##  [9] "quadrat"                         "transect"
## [11] "expunit"                         "nd"
## [13] "stationary"                      "permtrans"
## [15] "radial_transect"                 "circular_transect"
## [17] "roaming_transect"                "random_swim"
## [19] "lit"
```

Calculate # of sampling unit per record

Create a column with the sum of sampling units values to see how many different sampling units were adopted by each study

```
sampling_df %<>%
   mutate(type_per_record = rowSums(across(quadrat:lit)
                        )
   )
```

Calculate number of records by number of sampling units in %

```
sampling_df$type_per_record <- as.factor(sampling_df$type_per_record)

sampling_unit_per_record <- sampling_df %>%
  group_by(type_per_record) %>%
  summarise(n_records = length(type_per_record)) %>%
  mutate(percentage= round(n_records / sum(n_records) * 100,0))

sampling_unit_per_record
```

```
## # A tibble: 4 × 3
##   type_per_record n_records percentage
##   <fct>               <int>      <dbl>
## 1 0                      84         16
## 2 1                     398         76
## 3 2                      39          7
## 4 3                       1          0
```

Create a data frame with number of publication per sampling unit type

```
sampling_unit_tot <- sampling_df  %>%
  summarise(across(c(8:(length(sampling_df)-1)),
                   list(sum)))

sampling_unit_tot <- reshape2::melt(sampling_unit_tot)
```

```
## No id variables; using all as measure variables
```

```
colnames(sampling_unit_tot) <- c("Sampling_unit", "Count")

sampling_unit_tot %<>%
        mutate_at("Sampling_unit",
                  stringr::str_replace, "_1", "")
```

Simplified sampling units

Create a column with a simplified sampling units for transect and quadrat

```
  sampling_unit_tot %<>%
      mutate(Sampling_unit_macro = case_when(Sampling_unit == 'quadrat' |
                                  Sampling_unit == "permquad" ~ "Quadrat",
                                  Sampling_unit == "transect"|
                                  Sampling_unit == "lit" |
                                  Sampling_unit == "pit" |
                                  Sampling_unit == "permlit" |
                                  Sampling_unit == "permtrans" |
                                  Sampling_unit == "circular_transect" |
                                  Sampling_unit == "radial_transect"~ "Transe
ct",
                                  Sampling_unit == "nd" ~ "ND",
                                  Sampling_unit == "stationary" ~ "Stationary
",
                                  Sampling_unit == "expunit" ~ "Experimental 
unit",
                                  Sampling_unit == "sample" ~ "Sample",
                                  .default = "Other")
              )

# Change columns order
sampling_unit_tot <- cbind(sampling_unit_tot$Sampling_unit_macro,
```

```
                        sampling_unit_tot$Sampling_unit,
                        sampling_unit_tot$Count)

colnames(sampling_unit_tot) <- c("Sampling_unit_macro","Sampling_unit","Count
")

sampling_unit_tot <- as.data.frame(sampling_unit_tot)

sampling_unit_tot[,1:2] <- lapply(sampling_unit_tot[,1:2], as.factor)
sampling_unit_tot$Count <- as.numeric(sampling_unit_tot$Count)

sampling_unit_simplified <- aggregate(Count ~ Sampling_unit_macro, data = sam
pling_unit_tot, FUN = sum)

sampling_unit_simplified %<>%
  group_by(Sampling_unit_macro) %>%
  mutate(Percentage_macro= round(Count / sum(sampling_unit_simplified$Count)
* 100,1))

sampling_unit_simplified

## # A tibble: 7 × 3
## # Groups:   Sampling_unit_macro [7]
##   Sampling_unit_macro Count Percentage_macro
##   <fct>               <dbl>            <dbl>
## 1 Experimental unit      21              3.4
## 2 ND                     23              3.8
## 3 Other                   3              0.5
## 4 Quadrat               201             32.9
## 5 Sample                132             21.6
## 6 Stationary              8              1.3
## 7 Transect              223             36.5
```

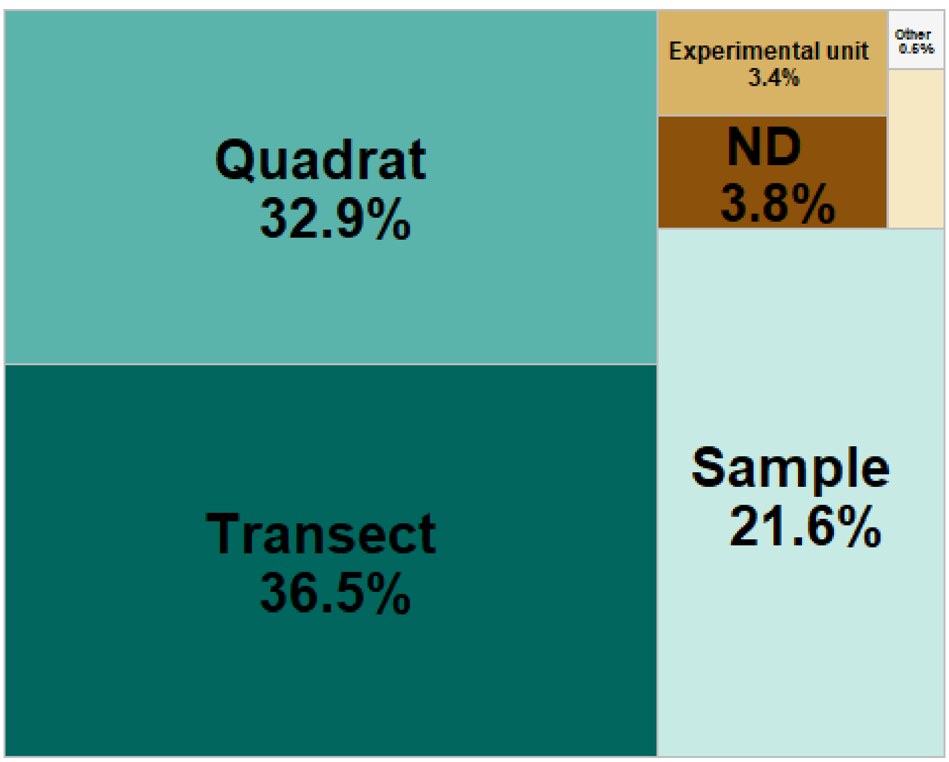
Build a treemap with simplified sampling unit

```
library(treemapify)

RdYlBu <- RColorBrewer::brewer.pal(7, "RdYlBu")
RdYlBu

## [1] "#D73027" "#FC8D59" "#FEE090" "#FFFFBF" "#E0F3F8" "#91BFDB" "#4575B4"

sampling_simplified_treemap <-
ggplot(sampling_unit_simplified,
       aes(area=Count,
           fill=Sampling_unit_macro,
           line = "black",
           label=paste0(Sampling_unit_macro, " \n ",Percentage_macro,"%"))
       )+
  treemapify::geom_treemap(layout = "squarified")+
```

```
  geom_treemap_text(place = "centre",
                    size = 20)+
    theme(plot.title = element_text(hjust = 0.5, size= 20),
          legend.position = "none")+
  scale_fill_manual(values= c("#FC8D59", "#FEE090","#FFFFBF", "#E0F3F8", "#D7
3027", "#91BFDB", "#4575B4"))
  scale_fill_brewer(palette = "RdYlBu")
```

```
## <ggproto object: Class ScaleDiscrete, Scale, gg>
##     aesthetics: fill
##     axis_order: function
##     break_info: function
##     break_positions: function
##     breaks: waiver
##     call: call
##     clone: function
##     dimension: function
##     drop: TRUE
##     expand: waiver
##     fallback_palette: function
##     get_breaks: function
##     get_breaks_minor: function
##     get_labels: function
##     get_limits: function
##     get_transformation: function
##     guide: legend
##     is_discrete: function
##     is_empty: function
##     labels: waiver
##     limits: NULL
##     make_sec_title: function
##     make_title: function
##     map: function
##     map_df: function
##     minor_breaks: waiver
##     n.breaks.cache: NULL
##     na.translate: TRUE
##     na.value: NA
##     name: waiver
##     palette: function
##     palette.cache: NULL
##     position: left
##     range: environment
##     rescale: function
##     reset: function
##     train: function
##     train_df: function
##     transform: function
##     transform_df: function
##     super:  <ggproto object: Class ScaleDiscrete, Scale, gg>
```

```r
sampling_unit_simplified %<>%
  mutate(case_when(Sampling_unit_macro == "Experimental unit" ~
          "Experimental \n unit", T ~ Sampling_unit_macro))

sampling_simplified_treemap <-
ggplot(sampling_unit_simplified,
       aes(area=Count,
           fill= reorder(Sampling_unit_macro, -Count),
           line = "black",
           label=paste0(Sampling_unit_macro, " \n ",Percentage_macro,"%"))
       )+
  treemapify::geom_treemap(layout = "squarified")+
  geom_treemap_text(place = "centre", fontface = "bold",
                    size = 20)+
    theme(plot.title = element_text(hjust = 0.5, size= 20),
          legend.position = "none")+
  scale_fill_manual(values=c("#01665E","#5AB4AC","#C7EAE5", "#8C510A","#D8B36
5","#F6E8C3","#F5F5F5")) # selected from "BrBG" palette of RBrewer

sampling_simplified_treemap
```



Export sampling_simplified_treemap

```r
ggsave("sampling_simplified_treemap.png",
       plot = sampling_simplified_treemap,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",
```

```
                    "temperate",
                    "Output-Visualization"))
```

Export sampling_unit_tot table for supplementary materials

```
library(openxlsx)

openxlsx::write.xlsx(sampling_unit_tot,
           file = here("Extracted_data_analysis",                        "tem
perate",                        "Output-Transformed_data",
                    "sampling_unit_tot.xlsx"))
```

Quadrat extracted from transect

```
print(paste("Number of publications where quadrat is the sampling unit or one
 of the sampling unit: ",
            nrow(quad_design <- subset(temperate,grepl("quad", samplingUnit)
                                       )
                            )
            )
)

## [1] "Number of publications where quadrat is the sampling unit or one of t
he sampling unit:  201"

print(paste("Number of publications where quadrats were collected along trans
ects: ", nrow(subset(quad_design,
            grepl("transect\\(quadrat", meso_design) |
              grepl("transect\\(depth\\(quadrat", meso_design)
            )
     )
)
)

## [1] "Number of publications where quadrats were collected along transects:
  126"
```

DATA TYPE

```
data_type <- count_results[["collectedDataType"]]

data_type$Percentage <- round(data_type$Percentage, 1)

data_type

## # A tibble: 33 × 3
##    collectedDataType              Count Percentage
##    <fct>                          <int>      <dbl>
##  1 nd                                 1        0.2
##  2 photo                             69       12.7
```

```
##   3 photo+sediment+video              2        0.4
##   4 photo+specimen                   41        7.6
##   5 photo+specimen+sediment+visual    2        0.4
##   6 photo+specimen+video             18        3.3
##   7 photo+specimen+video+visual       2        0.4
##   8 photo+specimen+video+water        1        0.2
##   9 photo+specimen+visual            10        1.9
##  10 photo+specimen+water              1        0.2
## # ℹ 23 more rows
```

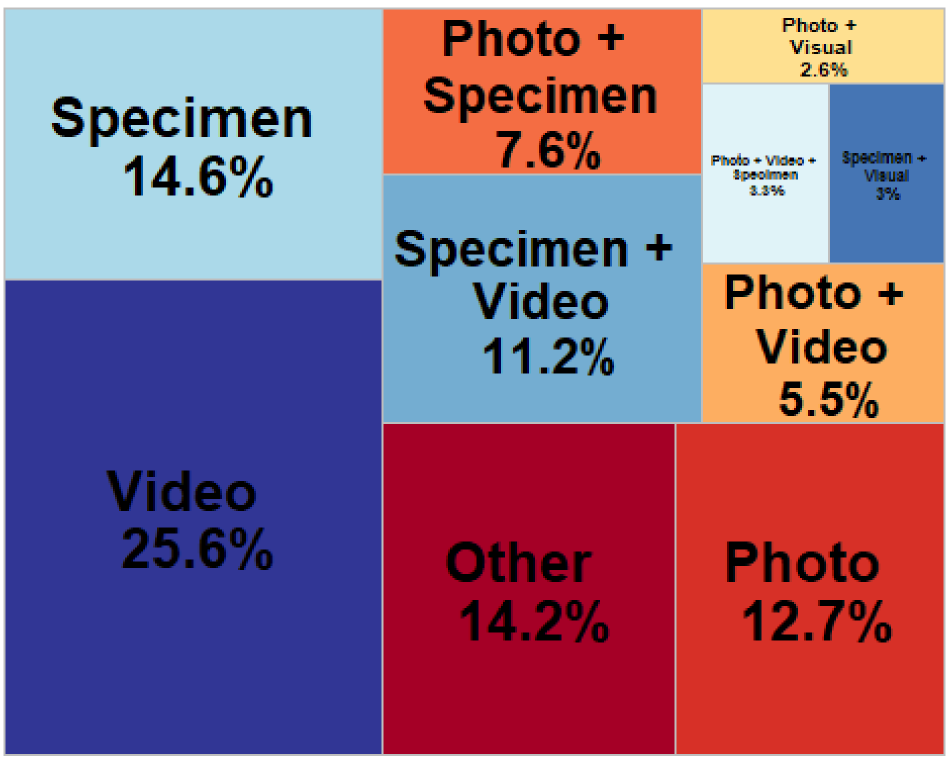
Lump uncommon combination into "Other"

```
data_type_simplified <- data_type %>%
  rename(`Data type`= collectedDataType) %>%
  mutate(`Data type` = as.character(`Data type`)) %>%
  filter(Percentage > 2) %>%
  bind_rows(
    data_type %>%
      filter(Percentage <= 2) %>%
      summarise(
        `Data type` = "other",
        Count = sum(.data$Count),
        Percentage = sum(.data$Percentage)
      )
  )


data_type_simplified %<>%
       mutate(`Data type` =
                case_when(`Data type` == 'photo'  ~ "Photo",
                          `Data type` == "video"~ "Video",
                          `Data type` == "other" ~ "Other",
                          `Data type` == "specimen" ~ "Specimen",
                          `Data type` == "photo+specimen" ~ "Photo + \n Speci
men",
                          `Data type` == "photo+specimen+video" ~ "Photo + Vi
deo + \n Specimen",
                          `Data type` == "photo+video" ~ "Photo + \n Video",
                          `Data type` == "photo+visual" ~ "Photo + \n Visual",
                          `Data type` == "specimen+visual" ~ "Specimen + \n V
isual",
                          `Data type` == "specimen+video" ~ "Specimen + \n Vi
deo",
                                  .default = "Other")
              )
```

Build a treemap with data type

```
library(treemapify)
library(RColorBrewer)
```

```
data_type_treemap <-
ggplot(data_type_simplified,
       aes(area=Count,
           fill=`Data type`,
           line = "black",
           label=paste0(`Data type`, " \n ",Percentage,"%"))
       )+
  treemapify::geom_treemap(layout = "squarified")+
  geom_treemap_text(place = "centre", fontface = "bold",
                    size = 20)+
    theme(plot.title = element_text(hjust = 0.5, size= 20),
          legend.position = "none")+
    scale_fill_brewer(palette = "RdYlBu")

data_type_treemap
```



Export sampling_simplified_treemap

```
ggsave("data_type_treemap.png",
       plot = data_type_treemap,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                     "temper
ate",                            "Output-Visualization"))
```

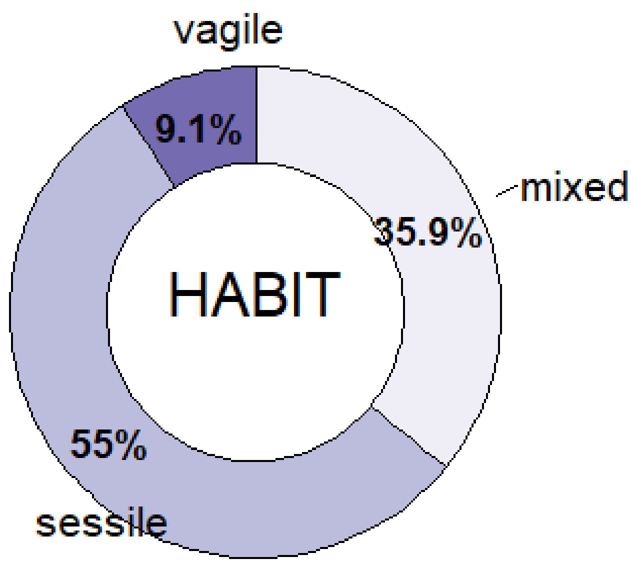
HABIT

```
habit <- as.data.frame(count_results["habit"])

# Rename columns
habit %<>%
  rename(Habit = habit.habit,
         Count = habit.Count,
         Percentage = habit.Percentage)
```

Make the habit donut plot

```
habit %<>%
  mutate(ymax = cumsum(Count),
    ymin = c(0, head(ymax, n = -1)),
    mid = (ymin + ymax) / 2,
    angle = 90 - (mid / sum(Count)) * 360
  )

# Split and assign labels
habit_inside <- habit %>%
  mutate(label = paste0(round(Percentage,1), "%"))

habit_outside <- habit %>%
  mutate(label = Habit)

# Plot
habit_donut <- ggplot() +
  # Donut segments
  geom_rect(data = habit,
            aes(ymin = ymin, ymax = ymax, xmin = 2.5, xmax = 4, fill = Habit),
            color = "black") +

  # Inside labels
  geom_text(data = habit_inside,
            aes(x = 3.2, y = mid, label = label),
            size = 5, color = "black",
            fontface = "bold") +

  # Outside labels with Habit name
  ggrepel::geom_text_repel(data = habit_outside,
                  aes(x = 4.3, y = mid, label = label),
                  size = 5.5,
                  nudge_x = 0.5,
                  direction = "y",
                  segment.color = "black",
                  hjust = 0,
                  box.padding = 0.3,
                  max.overlaps = Inf) +
```

```
  coord_polar(theta = "y") +
  xlim(0.2, 5) +
  scale_fill_brewer(palette = "Purples") +
  theme_void() +
  theme(legend.position = "none")+
  annotate(geom = 'text', x = 0.5, y = 0,
           label = c("HABIT"), size = 8)

habit_donut
```



Export habit_donut

```
ggsave("habit_donut.png",
       plot = habit_donut,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                        "temper
ate",                            "Output-Visualization"))
```

SUBSTRATE

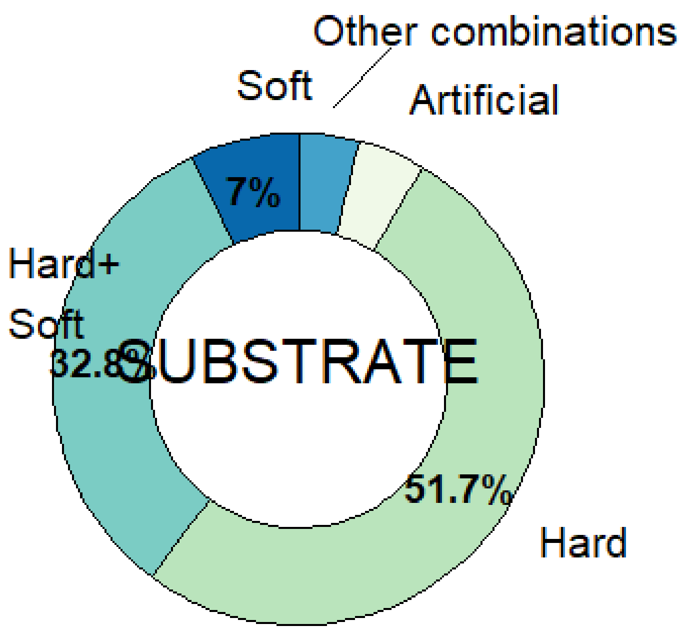
```
substrate <- as.data.frame(count_results["meso_substrate"])

# Rename columns
substrate %<>%
  rename(Substrate = meso_substrate.meso_substrate,
         Count = meso_substrate.Count,
```

```
        Percentage = meso_substrate.Percentage)

# Remove rows with nd
substrate <- substrate[!substrate$Substrate %in% "nd",]

# Merge substrate combination with less than 3% and recalculate Count and Per
centages
substrate <- substrate %>%
  mutate(Substrate = if_else(Percentage < 3, "Other combinations", Substrate))
 %>%
  group_by(Substrate) %>%
  select(-Percentage) %>%
  summarise(
    Count = sum(Count)) %>%
  mutate(Percentage = round(Count / sum(Count) * 100,1)
  )

substrate %<>%
  mutate(
    Substrate = case_when(Substrate == "hard" ~ "Hard",
                          Substrate == "soft"~ "Soft",
                          Substrate == "mixed" ~ paste0("Hard+","\n","Soft"),
                          Substrate == "artificial" ~ "Artificial",
                          Substrate == "Other combinations" ~ "Other combinat
ions"))
```

Make the substrate donut plot

```
substrate %<>%
  mutate(
    ymax = cumsum(Count),
    ymin = c(0, head(ymax, n = -1)),
    mid = (ymin + ymax) / 2,
    angle = 90 - (mid / sum(Count)) * 360
  )


# Split and assign labels
substrate_inside <- substrate %>%
    filter(Percentage >= 5) %>%
  mutate(label = paste0(Percentage, "%"))

substrate_outside <- substrate %>%
  mutate(label = Substrate)

# Plot
substrate_donut <- ggplot() +
  # Donut segments
  geom_rect(data = substrate,
```

```r
            aes(ymin = ymin, ymax = ymax, xmin = 2.5, xmax = 4, fill = Substr
ate),
            color = "black") +

  # Inside labels
  geom_text(data = substrate_inside,
            aes(x = 3.2, y = mid, label = label),
            size = 5, color = "black",
            fontface = "bold") +

  # Outside labels with substrate name
  ggrepel::geom_text_repel(data = substrate_outside,
                  aes(x = 4.4, y = mid, label = label),
                  size = 5.5,
                  nudge_x = 0.3,
                  direction = "y",
                  segment.color = "black",
                  hjust = 0,
                  box.padding = 0.3,
                  max.overlaps = Inf) +
  coord_polar(theta = "y") +
  xlim(0.2, 5) +
  scale_fill_brewer(palette = "GnBu") +
  theme_void() +
  theme(legend.position = "none")+
  annotate(geom = 'text', x = 0.5, y = 0,
           label = c("SUBSTRATE"), size = 8)

substrate_donut
```



Export the substrate donut plot

```
ggsave("substrate_donut.png",
       plot = substrate_donut,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                        "temper
ate",                             "Output-Visualization"))
```

BENTHIC POSITION

Create a data frame with benthic position

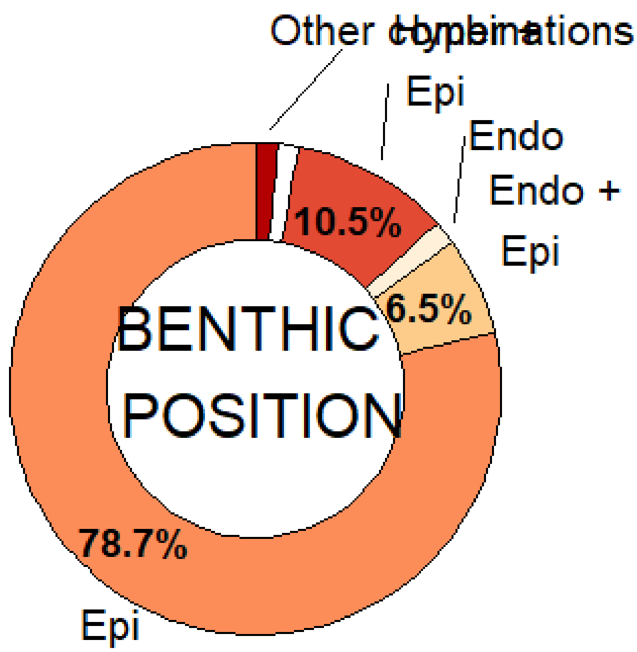
```
position <- as.data.frame(count_results["environmentalPosition"])

# Rename columns
position %<>%
  rename(Position = environmentalPosition.environmentalPosition,
         Count = environmentalPosition.Count,
         Percentage = environmentalPosition.Percentage)

# Merge position combination with less than 3% and recalculate Count and Perc
entages
position <- position %>%
  mutate(Position = if_else(Percentage < 1, "Other combinations", Position))
 %>%
  group_by(Position) %>%
```

```
  select(-Percentage) %>%
  summarise(
    Count = sum(Count)) %>%
  mutate(Percentage = round(Count / sum(Count) * 100,1)
  )
```

Make the benthic position donut plot

```
position %<>%
  mutate(
    Position = case_when(Position == "epi" ~ "Epi",
                         Position == "endo"~ "Endo",
                         Position == "dem+epi" ~ "Hyper + \n Epi",
                         Position == "endo+epi" ~ "Endo + \n Epi",
                         Position == "Other combinations" ~ "Other combinati
ons"),
    ymax = cumsum(Count),
    ymin = c(0, head(ymax, n = -1)),
    mid = (ymin + ymax) / 2,
    angle = 90 - (mid / sum(Count)) * 360
  )


# Split and assign labels
position_inside <- position %>%
  filter(Percentage > 3) %>%
  mutate(label = paste0(Percentage, "%"))

position_outside <- position %>%
  mutate(label = Position)

# Plot
benthic_position_donut <- ggplot() +
  # Donut segments
  geom_rect(data = position,
            aes(ymin = ymin, ymax = ymax, xmin = 2.5, xmax = 4, fill = Positi
on),
            color = "black") +

  # Inside labels
  geom_text(data = position_inside,
            aes(x = 3.2, y = mid, label = label),
            size = 5, color = "black",
            fontface = "bold") +

  # Outside labels with Position name
  ggrepel::geom_text_repel(data = position_outside,
                  aes(x = 4.2, y = mid, label = label),
                  size = 5.5,
```

```
                    nudge_x = 0.3,
                    direction = "y",
                    segment.color = "black",
                    hjust = 0,
                    box.padding = 0.3,
                    max.overlaps = Inf) +
  coord_polar(theta = "y") +
  xlim(0.2, 5) +
  scale_fill_brewer(palette = "OrRd") +
  theme_void() +
  theme(legend.position = "none")+
  annotate(geom = 'text', x = 0.5, y = 0,
           label = c("BENTHIC \n POSITION"), size = 8)

benthic_position_donut

## Warning: Removed 1 row containing missing values or values outside the sca
le range
## (`geom_text_repel()`).
```



Export the benthic_position donut plot

```
ggsave("benthic_position_donut.png",
       plot = benthic_position_donut,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                    "temper
ate",                             "Output-Visualization"))
```

```
## Warning: Removed 1 row containing missing values or values outside the sca
le range
## (`geom_text_repel()`).
```

TARGET

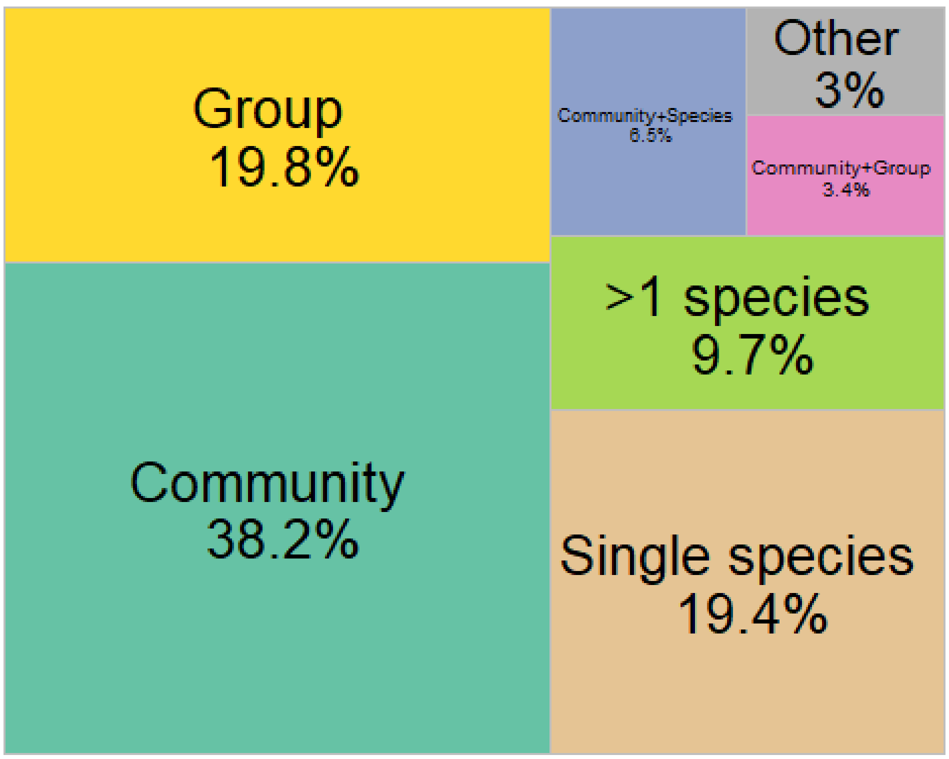
Target level

```
target_level <- as.data.frame(count_results["taxonomicTargetLevel"])

# Rename columns
target_level %<>%
  rename(Level = taxonomicTargetLevel.taxonomicTargetLevel,
         Count = taxonomicTargetLevel.Count,
         Percentage = taxonomicTargetLevel.Percentage)

# Remove rows with nd
target_level <- target_level[!target_level$Level %in% "species,species",]
target_level <- target_level[!target_level$Level %in% ",species",]

# Merge habit combination with less than 3% and recalculate Count and Percent
ages
target_level <- target_level %>%
  mutate(Level = if_else(Percentage < 3, "Other combinations", Level)) %>%
  group_by(Level) %>%
  select(-Percentage) %>%
  summarise(
    Count = sum(Count)) %>%
  mutate(Percentage = round(Count / sum(Count) * 100,1)
  )
```

Treemap with the taxonomic target level

```
target_level_labels <- target_level %<>%
  mutate(
    Level = case_when(Level == "comm" ~ "Community",
                      Level == "group"~ "Group",
                      Level == "single" ~ "Single species",
                      Level == "species" ~ ">1 species",
                      Level == "comm+group" ~ "Community+Group",
                      Level == "comm+species" ~ "Community+Species",
                      Level == "group+species" ~ "Group+\nSpecies",
                      Level == "Other combinations" ~ "Other")
    )

target_level_treemap <- ggplot(target_level_labels,
       aes(area=Count,
           fill=Level,
           line = "black",
```

```
          label=paste0(Level,
                          " \n ",Percentage,"%"))
     )+
  treemapify::geom_treemap(layout = "squarified")+
  treemapify::geom_treemap_text(place = "centre",
                        size = 20)+
    theme(plot.title = element_text(hjust = 0.5, size= 20),
          legend.position = "none")+
  scale_fill_manual(values= c("#A6D854" ,"#66C2A5","#E78AC3", "#8DA0CB","#FFD
92F","#B3B3B3", "#E5C494", "#FC8D62"))

target_level_treemap
```




```
ggsave("target_level_treemap.png",
       plot = target_level_treemap,
       units = "in", width = 9, height = 6,
       path = here ("Extracted_data_analysis",                        "temper
ate",                        "Output-Visualization"))
```

Detected phyla in community studies

```
detected <- temperate %>%
    filter(str_detect(taxonomicTargetLevel,"comm")) %>%
  select(File,
         detected_meso_cnidaria,
         detected_meso_sponges,
         detected_meso_seagrass,
```

```
         detected_meso_algae,
         detected_meso_mollusca,
         detected_meso_chondrichthyes,
         detected_meso_fish,
         detected_meso_crustacea,
         detected_meso_echino,
         detected_meso_other_taxa)

detected_lvls <-
  unique(
    unlist(
      str_split(detected$detected_meso_other_taxa, "\\+")
                                   )
                        )
detected_lvls<- detected_lvls[! detected_lvls %in% c("",NA,"notapp",",","nd")]
detected[,detected_lvls] <- sapply(detected_lvls,
                                      function(y){
                                        str_count(detected$detected_meso_othe
r_taxa, y)
                                        }
                                      )

rm(detected_lvls)

detected %<>%
  mutate(across(everything(), as.character)) %>%
  replace(is.na(.), "0") %>%
  select(-detected_meso_other_taxa) %>%
  filter(if_all(everything(), ~ !grepl("notapp", .x)))

detected <- detected %>%
  mutate(across(-File, ~ case_when(
    .x %in% c("x", "x,", "xx") ~ "1",
    .x == "," ~ "0",
    TRUE ~ .x
  ))) %>%
  mutate(across(-File, as.numeric))
```

Calculate number of publications per taxonomic group

```
detected_tot <-
  detected %>%
  summarise(across(where(is.numeric),
                   list(sum)
                   )
            )

detected_tot  <- reshape2::melt(detected_tot, id.vars = NULL)

colnames(detected_tot) <- c("Taxa", "Count")
```

```
detected_tot %<>%
        mutate_at("Taxa",
                  stringr::str_replace, "_1", "") %>%
        mutate_at("Taxa",
                  stringr::str_replace, "detected_meso_", "")


detected_tot %<>%
      mutate(Percentage= round(Count / nrow(detected) * 100,1))

rm(detected)
```

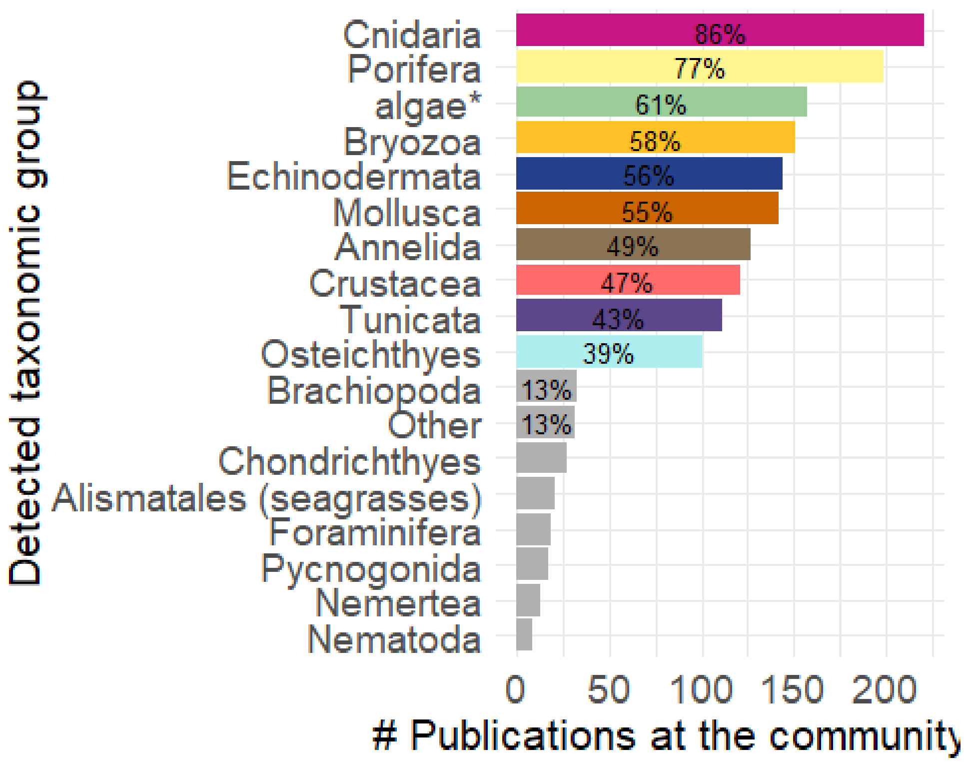
Merge poorly represented taxonomic groups into “Other”

```
detected_tot_collapsed <- detected_tot %>%
  mutate(Taxa = if_else(Percentage < 3, "Other", Taxa)) %>%
  group_by(Taxa) %>%
  summarise(
    Count = sum(Count),
    Percentage = sum(Percentage)
  )
```

Create a bar chart with detected taxonomic groups

```
detected_bar <- ggplot(detected_tot_collapsed, aes(x = reorder(Taxa, Count),
y = Count, fill= Taxa)) +
  geom_bar(stat = "identity") +
  geom_text(
    data = detected_tot_collapsed %>% filter(Percentage > 11),
    aes(label = paste0(round(Percentage, 0), "%")),
    position = position_stack(vjust = 0.5),
    color = "black",
    size = 3.5
  )+
  labs(y = "# Publications at the community level", x = "Detected taxonomic g
roup") +
  theme_minimal() +
  coord_flip()+  # optional: flips for horizontal bars
  theme(legend.position ="none")+
  scale_x_discrete(labels = rev(c("Cnidaria",
                                "Porifera",
                                "algae*",
                                "Bryozoa",
                                "Echinodermata",
                                "Mollusca",
                                "Annelida",
                                "Crustacea",
                                "Tunicata",
                                "Osteichthyes",
```

```
                            "Brachiopoda",
                            "Other",
                            "Chondrichthyes",
                            "Alismatales (seagrasses)",
                            "Foraminifera",
                            "Pycnogonida",
                            "Nemertea",
                            "Nematoda")))

detected_palette <- c("algae" ="darkseagreen3",
                      "cnidaria" = "mediumvioletred",
                      "crustacea" = "indianred1",
                      "sponges" = "khaki1",
                      "mollusca" = "darkorange3",
                      "fish" = "paleturquoise2",
                      "echino" = "royalblue4",
                      "Tunicata" = "mediumpurple4",
                      "chondrichthyes" = "gray69",
                      "Brachiopoda" = "gray69",
                      "Other" = "gray69",
                      "Nemertea" = "gray69",
                      "Pycnogonida" = "gray69",
                      "seagrass" =  "gray69",
                      "Nematoda" = "gray69",
                      "Foraminifera"   =    "gray69",
                      "Ciliata" = "gray69",
                      "Bryozoa" = "goldenrod1",
                      "Annelida" = "burlywood4")



detected_bar +
  scale_fill_manual(values = detected_palette,
    limits = names(detected_palette))+
  theme(axis.text = element_text(size = 14),
        axis.title = element_text(size = 16)
        )
```




```
ggsave("detected_bar.png",
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                        "temper
ate",                             "Output-Visualization"))
```

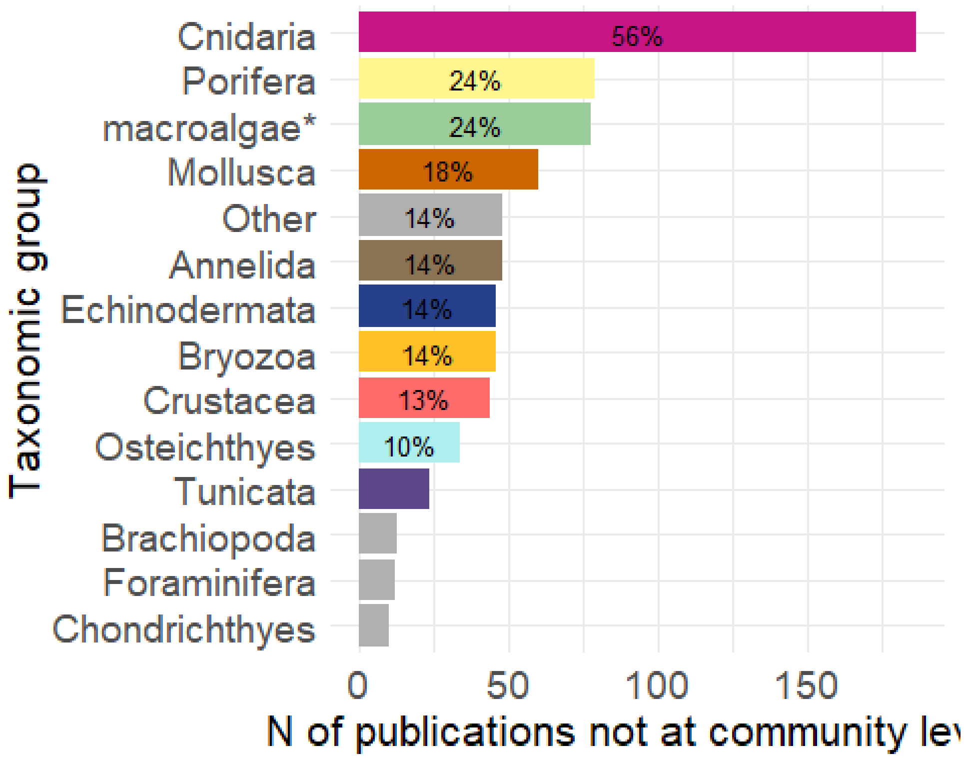
Target taxonomic groups not at the community level

```
group <- temperate %>%
    filter(taxonomicTargetLevel != "comm") %>%
  select(File,
         detected_meso_cnidaria,
         detected_meso_sponges,
         detected_meso_seagrass,
         detected_meso_algae,
         detected_meso_mollusca,
         detected_meso_chondrichthyes,
         detected_meso_fish,
         detected_meso_crustacea,
         detected_meso_echino,
         detected_meso_other_taxa,
         targetTaxa)

group_lvls <-
  unique(
    unlist(
      str_split(group$detected_meso_other_taxa, "\\+")
                                 )
```

```
                         )
group_lvls<- group_lvls[! group_lvls %in% c("",NA,"notapp",",","nd")]
group[,group_lvls] <- sapply(group_lvls,
                                      function(y){
                                        str_count(group$detected_meso_other_t
axa, y)
                                        }
                                      )

rm(group_lvls)

group <- group %>%
  mutate(across(everything(), as.character)) %>%
  replace(is.na(.), "0") %>%
  select(-detected_meso_other_taxa,-targetTaxa) %>%
  filter(if_all(everything(), ~ !grepl("notapp", .x))) %>%
  mutate(across(-File, ~ case_when(
    .x %in% c("x", "x,", "xx") ~ "1",
    .x == "," ~ "0",
    TRUE ~ .x
  ))) %>%
  mutate(across(-File, as.numeric))

group_tot <-
  group %>%
  summarise(across(where(is.numeric),
                   list(sum)
                   )
            )

group_tot  <- reshape2::melt(group_tot, id.vars = NULL)

colnames(group_tot) <- c("Taxa", "Count")

group_tot %<>%
        mutate_at("Taxa",
                  stringr::str_replace, "_1", "") %>%
        mutate_at("Taxa",
                  stringr::str_replace, "detected_meso_", "")



group_tot %<>%
      mutate(Percentage= round(Count / nrow(group) * 100,1))

group_tot_collapsed <- group_tot %>%
  mutate(Taxa = if_else(Percentage < 3, "Other", Taxa)) %>%
  group_by(Taxa) %>%
  summarise(
    Count = sum(Count),
    Percentage = sum(Percentage),
```

```
    .groups = "drop"
  )

group_tot_collapsed %<>%
  mutate(Taxa = case_when(Taxa == 'echino' ~ "Echinodermata",
                          Taxa == 'fish' ~ "Osteichthyes",
                          Taxa == 'mollusca' ~ "Mollusca",
                          Taxa == 'algae' ~ "macroalgae*",
                          Taxa == "chondrichthyes" ~ "Chondrichthyes",
                          Taxa == "sponges" ~ "Porifera",
                          Taxa == "crustacea" ~ "Crustacea",
                          Taxa == "cnidaria" ~ "Cnidaria",
                          Taxa == "algae" ~ "algae*",
         .default = Taxa))

group_bar <- ggplot(group_tot_collapsed, aes(x = reorder(Taxa, Count), y = Co
unt, fill= Taxa)) +
  geom_bar(stat = "identity") +
  geom_text(
    data = group_tot_collapsed %>% filter(Percentage > 10),
    aes(label = paste0(round(Percentage, 0), "%")),
    position = position_stack(vjust = 0.5),
    color = "black",
    size = 3.5
  )+
  labs(y = "N of publications not at community level", x = "Taxonomic group")
 +
  theme_minimal() +
  coord_flip()+  # optional: flips for horizontal bars
  theme(legend.position ="none")

group_palette <- c("macroalgae*" ="darkseagreen3",
                      "Cnidaria" = "mediumvioletred",
                      "Crustacea" = "indianred1",
                      "Porifera" = "khaki1",
                      "Mollusca" = "darkorange3",
                      "Osteichthyes" = "paleturquoise2",
                      "Echinodermata" = "royalblue4",
                      "Tunicata" = "mediumpurple4",
                      "Chondrichthyes" = "gray69",
                      "Other" = "gray69",
                      "Brachiopoda" =  "gray69",
                      "Foraminifera" =  "gray69",
                      "Bryozoa" = "goldenrod1",
                      "Annelida" = "burlywood4")


group_bar <- group_bar +
  scale_fill_manual(values = group_palette,
    limits = names(group_palette))+
    theme(axis.text = element_text(size = 14),
```

```
        axis.title = element_text(size = 16)
        )

group_bar
```



Export bar plot

```
ggsave("group_bar.png",
       plot = group_bar,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                        "temper
ate",                             "Output-Visualization"))
```

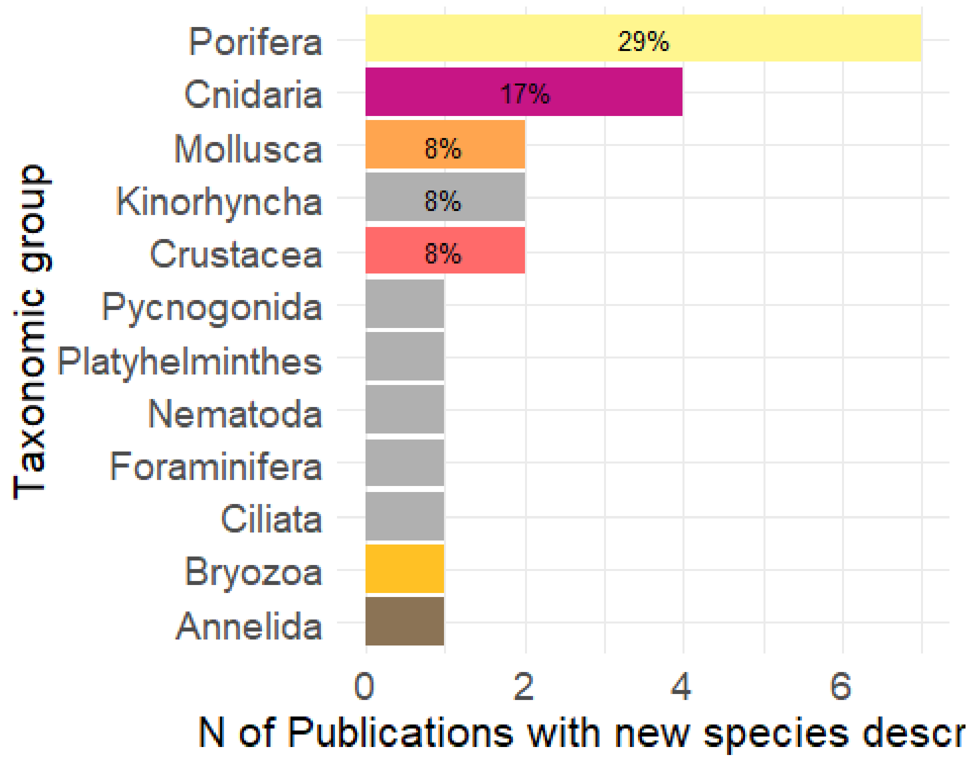
New species described per phyla

```
new_spec <- temperate %>%
  filter(str_detect(taxonomic_publication, "taxa"))

new_spec %<>%
  select(File,
         detected_meso_cnidaria,
         detected_meso_sponges,
         detected_meso_seagrass,
         detected_meso_algae,
         detected_meso_mollusca,
         detected_meso_chondrichthyes,
         detected_meso_fish,
         detected_meso_crustacea,
```

```
         detected_meso_echino,
         detected_meso_other_taxa)

new_spec_lvls <-
  unique(
    unlist(
      str_split(new_spec$detected_meso_other_taxa, "\\+")
                                )
                         )
new_spec_lvls<- new_spec_lvls[! new_spec_lvls %in% c("",NA,"notapp")]
new_spec[,new_spec_lvls] <- sapply(new_spec_lvls,
                                      function(y){
                                        str_count(new_spec$detected_meso_othe
r_taxa, y)
                                        }
                                      )

rm(new_spec_lvls)

new_spec  %<>%
  mutate(across(everything(), as.character)) %>%
  replace(is.na(.), "0") %>%
  select(-detected_meso_other_taxa) %>%
  filter(if_all(everything(), ~ !grepl("notapp", .x))) %>%
  mutate(across(-File, ~ case_when(
    .x %in% c("x") ~ "1",
    TRUE ~ .x
  ))) %>%
  mutate(across(-File, as.numeric))

new_spec_tot <-
  new_spec %>%
  summarise(across(where(is.numeric),
                   list(sum)
                   )
            )

rm(new_spec)
new_spec_tot  <- reshape2::melt(new_spec_tot, id.vars = NULL)

colnames(new_spec_tot) <- c("Taxa", "Count")

new_spec_tot %<>%
        mutate_at("Taxa",
                  stringr::str_replace, "_1", "") %>%
          mutate_at("Taxa",
                  stringr::str_replace, "detected_meso_", "")


new_spec_tot %<>%
```

```r
  filter(Count>0) %>%
      mutate(Percentage= round(Count / sum(Count) * 100,1))

new_spec_plot <-   ggplot(new_spec_tot,
                          aes(x = reorder(Taxa, Count), y = Count,
                              fill= Taxa)) +
  geom_bar(stat = "identity") +
  geom_text(

    data = new_spec_tot %>% filter(Percentage > 8),
    aes(label = paste0(round(Percentage, 0), "%")),
    position = position_stack(vjust = 0.5),
    color = "black",
    size = 3.5
  )+
  labs(y = "N of Publications with new species descriptions", x = "Taxonomic
group") +
  theme_minimal() +
  coord_flip()+  # optional: flips for horizontal bars
  theme(legend.position ="none")+
  scale_x_discrete(labels = rev(c("Porifera",
                                "Cnidaria",
                                "Mollusca",
                                "Kinorhyncha",
                                "Crustacea",
                            "Pycnogonida",
                            "Platyhelminthes",
                            "Nematoda",
                            "Foraminifera",
                            "Ciliata",
                            "Bryozoa",
                            "Annelida"))
                   )

new_spec_palette <- c("cnidaria" = "mediumvioletred",
                      "crustacea" = "indianred1",
                      "sponges" = "khaki1",
                      "mollusca" = "tan1",
                      "Kinorhyncha" = "gray69",
                      "Pycnogonida" = "gray69",
                      "Platyhelminthes" =  "gray69",
                      "Nematoda" = "gray69",
                      "Foraminifera"   =    "gray69",
                      "Ciliata" = "gray69",
                      "Bryozoa" = "goldenrod1",
                      "Annelida" = "burlywood4")

new_species_bar <- new_spec_plot +
  scale_fill_manual(values = new_spec_palette,
    limits = names(new_spec_palette))+
```

```
    theme(axis.text = element_text(size = 14),
        axis.title = element_text(size = 16)
        )

new_species_bar
```



Export bar plot

```
ggsave("new_species_bar.png",
       plot = new_species_bar,
       units = "in", width = 7, height = 6,
       path = here ("Extracted_data_analysis",                         "temper
ate",                             "Output-Visualization"))
```

New species descriptions that included molecular analyses

```
new_spec_molecular <- temperate %>%
  filter(str_detect(taxonomic_publication, "taxa") &
           molecularAnalyses != "no" &
           molecularAnalyses != "nd" &
           molecularAnalyses != "notapp")

nrow(new_spec_molecular)

## [1] 6
```

DATA CROSSING

```
table(temperate$meso_protection)

##
## mixed    nd    no   yes
##    82   291    57   112
```

VISUALLY SURVEYED SURFACE

```
temperate$visuallySurveyedSurface <- as.factor(temperate$visuallySurveyedSurf
ace)
summary(subset(temperate, is.na(temperate$taxonomic_publication))$visuallySur
veyedSurface)

##          <1           >1          >10         >100        >1000         >10
000
##           9           23           54           76           83
 51
##      >100000     >1000000    >10000000   >100000000 >10000000000
 nd
##          16            4            1            1            1
179
##          no       notapp
##           2           26
```

Observation

```
print(paste("Number of manipulative experimental studies:",nrow(subset(temper
ate, grepl("exp", temperate$observationalORexperimental)))))

## [1] "Number of manipulative experimental studies: 33"

print(paste("Number of manipulative experimental studies in the field of rest
oration:", nrow(subset(temperate, grepl("exp", temperate$observationalORexper
imental) & temperate$field_rest != 0)

                  )
              )
      )

## [1] "Number of manipulative experimental studies in the field of restorati
on: 6"
```

Temp replicates & exp

```
print(paste("# Publications that included temporal replicates:", nrow(subset
(temperate, grepl("yes", temporalReplicates)))))

## [1] "# Publications that included temporal replicates: 136"
```

```r
print(paste("# Publications that included temporal replicates:", nrow(subset
(temperate, (grepl("exp", temperate$observationalORexperimental) & grepl("yes
", temporalReplicates))))))
```

```
## [1] "# Publications that included temporal replicates: 26"
```

Health

```r
health <- subset(temperate, grepl("Health", indicators) | grepl("Disease", in
dicators))

unique(health$targetTaxa)
```

```
##  [1] "Corallinales"

##  [2] "no"

##  [3] "Paramuricea_clavata"

##  [4] "Corallium_rubrum"

##  [5] "Bryozoa+Corallium_rubrum+Leptopsammia_pruvoti+Madracis_pharensis"

##  [6] "Eunicella_cavolini+Paramuricea_clavata"

##  [7] "Pachastrella monilifera+Poecillastra_compressa"

##  [8] "Peltaster_placenta+Peltaster_placenta"

##  [9] "Eunicella_verrucosa"

## [10] "Corallium_rubrum+Eunicella_cavolini"

## [11] "Adeonella_calveti+Myriapora_truncata+Pentapora_fascialis+Reteporella
_grimaldii+Smittina_cervicornis"
## [12] "Eunicella_cavolini+Eunicella singularis+Paramuricea_clavata"

## [13] "Cidaris_spp+Corallium_rubrum+Eunicella_cavolinii"

## [14] "Anthipathella_subpinnata+Eunicella_cavolini+Paramuricea_clavata"

## [15] "Viminella_flagellum"

## [16] "Eunicella_singularis"

## [17] "Porifera"

## [18] "Octocorallia"
```

```
## [19] "Anthozoa"

## [20] "notapp"

## [21] "Eunicella_cavolinii+Eunicella_singularis+Eunicella_verrucosa+Paramur
icea_clavata"
## [22] "Placopecten_magellanicus"

## [23] "Eunicella_cavolini+Eunicella_singularis+Paramuricea_clavata"

## [24] "Eunicella_singularis+Paramuricea_clavata"

## [25] "Balticina_willemoesi"

## [26] "Errina_novaezelandiae"

## [27] "Anthipatharia"

## [28] "Eunicella_cavolini"

## [29] "Acropora_austera+Anthozoa"

## [30] "Antipatharia+Octocorallia"

## [31] "Antipatharia+Octocorallia+Porifera"

## [32] "Spinimuricea_klavereni"

## [33] "Desmophyllum_pertusum"
```

ROV & sample

```
rov <- subset(temperate, grepl("rov", meso_samplingPlatform) )

rov_specimen <- subset(rov,  grepl("specimen", collectedDataType) )

print(paste("Percentage of publications relying solely on ROV where specimens
 were collected:", round(nrow(rov_specimen)/nrow(rov)*100,1),"%"))

## [1] "Percentage of publications relying solely on ROV where specimens were
 collected: 40.9 %"

rm(rov, rov_specimen)
```

Molecular techniques in taxonomic and non-taxonomic studies

```
temperate$taxonomic_publication <- as.factor(temperate$taxonomic_publication )
```

```
molecular <- temperate %>%
  filter(molecularAnalyses != "no" &
           molecularAnalyses != "notapp" &
           molecularAnalyses != "nd")

print(paste("# Publications that adopted molecular techniques: ",nrow(molecul
ar)))

## [1] "# Publications that adopted molecular techniques:  42"

molecular_notaxa <- molecular %>%
  filter(is.na(taxonomic_publication))

print(paste("# Publications that adopted molecular techniques not for taxonom
y: ",nrow(molecular_notaxa)))

## [1] "# Publications that adopted molecular techniques not for taxonomy:  3
6"

rm(molecular, molecular_notaxa)
```

SESSION INFORMATION AND REFERENCES

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=Italian_Italy.utf8  LC_CTYPE=Italian_Italy.utf8
## [3] LC_MONETARY=Italian_Italy.utf8 LC_NUMERIC=C
## [5] LC_TIME=Italian_Italy.utf8
##
## time zone: Europe/Rome
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] RColorBrewer_1.1-3 treemapify_2.6.0   openxlsx_4.2.8.1   tibble_3.3.1

##  [5] purrr_1.2.1        stringr_1.6.0      magrittr_2.0.4     readxl_1.4.5

##  [9] ggplot2_4.0.2      dplyr_1.2.0        here_1.0.2
##
## loaded via a namespace (and not attached):
##  [1] utf8_1.2.6        generics_0.1.4     tidyr_1.3.2       stringi_1.8.7
```

```
##  [5] digest_0.6.39     evaluate_1.0.5    grid_4.5.2        fastmap_1.2.0

##  [9] cellranger_1.1.0  rprojroot_2.1.1   plyr_1.8.9        ggrepel_0.9.6

## [13] zip_2.3.3         scales_1.4.0      textshaping_1.0.4 cli_3.6.5

## [17] rlang_1.1.7       withr_3.0.2       yaml_2.3.12       otel_0.2.0

## [21] tools_4.5.2       reshape2_1.4.5    forcats_1.0.1     vctrs_0.7.1

## [25] R6_2.6.1          lifecycle_1.0.5   ragg_1.5.0        pkgconfig_2.0.3

## [29] pillar_1.11.1     gtable_0.3.6      glue_1.8.0        Rcpp_1.1.1

## [33] ggfittext_0.10.3  systemfonts_1.3.1 xfun_0.56         tidyselect_1.2.
1
## [37] rstudioapi_0.18.0 knitr_1.51        farver_2.1.2      htmltools_0.5.9

## [41] rmarkdown_2.30    svglite_2.2.2     labeling_0.4.3    compiler_4.5.2

## [45] S7_0.2.1
```

```
library(report)
report::report(sessionInfo())
```

```
## Analyses were conducted using the R Statistical language (version 4.5.2; R
 Core
## Team, 2025) on Windows 11 x64 (build 26200), using the packages magrittr
## (version 2.0.4; Bache S, Wickham H, 2025), report (version 0.6.3; Makowski
 D et
## al., 2023), here (version 1.0.2; Müller K, 2025), tibble (version 3.3.1; M
üller
## K, Wickham H, 2026), RColorBrewer (version 1.1.3; Neuwirth E, 2022), openx
lsx
## (version 4.2.8.1; Schauberger P, Walker A, 2025), ggplot2 (version 4.0.2;
## Wickham H, 2016), stringr (version 1.6.0; Wickham H, 2025), readxl (versio
n
## 1.4.5; Wickham H, Bryan J, 2025), dplyr (version 1.2.0; Wickham H et al.,
## 2026), purrr (version 1.2.1; Wickham H, Henry L, 2026) and treemapify (ver
sion
## 2.6.0; Wilkins D, 2025).
##
## References
## ----------
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## <https://doi.org/10.32614/CRAN.package.magrittr>, R package version 2.0.4,
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```

```
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(2023).
## "Automated Results Reporting as a Practical Tool to Improve Reproducibilit
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## Methodological Best Practices Adoption." _CRAN_.
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## doi:10.32614/CRAN.package.purrr <https://doi.org/10.32614/CRAN.package.pur
rr>,
```

```
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0,
## <https://CRAN.R-project.org/package=treemapify>.
```